

ENGINEERING: AN INTRODUCTION FOR HIGH SCHOOL

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CK-12

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CHAPTER OVERVIEW

1: Introduction to Engineering

1.1: Introduction

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1.1: Introduction

Much of our modern society depends on engineered artifacts to function, but many members of modern society are not aware of the engineering techniques and practices that have developed the technology and infrastructure on which we rely. iPods, cell phones, airplanes, bridges, buildings, vehicles, computers, etc. are designed and created by engineers. This textbook introduces engineering techniques and practices to high school students. The goals of this book are to help students gain an appreciation for engineering and its role throughout human history, understand what engineers do, understand the skills and processes engineers bring to their work, and appreciate how the work of engineers shapes and is shaped by their society. The authors hope that this book may inspire students to pursue a career in engineering.

This book is a Flexbook® textbook—an open-source book developed with the support of and within the context of CK-12's mission; the Flexbook® format allows the book to be customized for multiple audiences. This engineering text is a living document that can be updated, expanded, and repurposed as necessary to support specific standards and classroom needs.

The text is written to meet draft ASEE K-12 standards for engineering. Each chapter corresponds to an outcome in the draft standard. While the standards have not yet been finalized and formally adopted, the Flexbook® format allows the text to evolve in response to changes in the standards, so that the text's content and structure will fully support them.

The text was collaboratively written by university engineering and education faculty members at Arizona State University. The text currently has four content chapters that cover the nature of engineering, engineering and society, engineering design, and the connection between engineering, science, and mathematics.

The authors are grateful to CK-12 for providing the infrastructure and support that has made this textbook possible. We see this book as a seed, and hope that it becomes a starting point on which others can build.

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CHAPTER OVERVIEW

2: Nature of Engineering

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2.1: About the Nature of Engineering

This chapter explores the nature of engineering. As you read this chapter, you will discover: what engineers do; some of the skills needed to be an engineer; various types of engineering careers and specializations; the educational requirements to be an engineer; licensure of engineers; the impact engineering has had on society; and some possible scenarios for the future of engineering.

Chapter Learning Objectives

After working through this chapter, you should be able to

- describe what engineers do,
- describe the education and skills necessary for engineering,
- describe the impact of engineering on society.

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2.2: Discovering Engineering

Who are engineers and what do they do? Why are the activities of engineers important? In this section, we will begin to discover some answers to these questions.

Some Practicing Engineers

Activity

What do you already know about engineering and engineers? Imagine an engineer at work. (You might want to get out a paper and something to write with.) What does the engineer look like? What is the engineer wearing? Where is the engineer working, and what are they doing? What does the engineer spend most of the day doing? What sorts of tools is the engineer using to help with their work? Is the engineer working alone or with others?

Capture your ideas by making a list of your answers to the questions above, or by drawing a picture of the engineer that you are imagining. When you are finished share your drawing or list with someone else. How are your lists or pictures similar? How are they different?

Continue imagining your engineer and add to your drawing or list. What sort of education does your engineer have? What sorts of classes did they take in college? What does the engineer do very well, and what does the engineer not do well? Does your engineer have an area of specialization? If so, what? How much money does your engineer make in a given year? Now share the expanded version of your engineer with someone else, and once again discuss the similarities and differences.

As we progress through the chapter, we will check to see if your ideas change.

Now that you have envisioned an engineer, let us look at some real-life profiles of practicing engineers. As you read each profile, note the attributes that you included in your picture or list and make a new list of the attributes that differ from your picture or list.

Profile 1. Ashley is in charge of product development and support for a large **electronics** product company in the Pacific Northwest. She manages two engineering teams. She is 39 years old and likes living in the Pacific Northwest because of the outdoor activities such as hiking and camping. The members of her engineering teams live in other cities and most of them live outside the United States in countries that include India, China, Sri Lanka, and Malaysia. Each location has some particular engineering advantage. For example, the United States is the best place to design products and manage product development and support; India has a very good system to support technology development and it is less expensive to develop software there; China was selected as the best place to manufacture computer chips; and Malaysia and Sri Lanka were selected to manufacture and assemble the rest of the products.



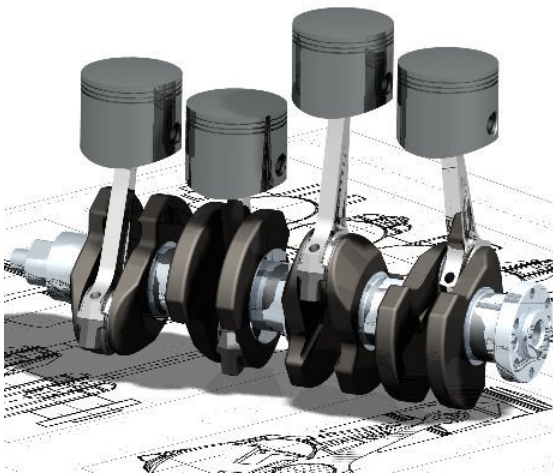
One of Ashley's projects might include developing components that will be part of a satellite antenna system such as this one at the Cryptologic Operations Center in Misawa, Japan.

Most days Ashley works out of her home. Because her engineering team members are located all over the world, she must be available to communicate with them 24 hours a day or whenever a problem arises. To aid this global communication, Ashley's computer sounds a bell any time one of her team members sends important email or needs to talk with her directly. Since she is available 24 hours a day, her daily routine is very flexible. Ashley can usually choose her own work schedule, except when she has scheduled meetings or urgent communication demands, which might be only two or three times each week. Sometimes she spends the morning working in her garden after handling some of her morning communication (the bell also rings outside), and she also takes a break to paint most afternoons. Ashley travels to each of the team member locations one or two times each year.

Ashley's most important tools are her computer and her mobile phone. She has excellent communication skills and knows how to relate to the different cultures of her team members. Ashley also has a broad knowledge of electronic product systems. Although she is not an expert in any of the individual components, she understands how each of the components works together (Figure above).

Ashley was in college for four and one-half years studying for her engineering degree. She spent the first year at a community college before transferring to a university. Ashley liked math in high school but did not settle on an engineering major until she was in her junior year of college and had to use math to analyze and design an electronics project. Her favorite courses were those that explained how electronic devices worked. Ashley earned \$90,000 in 2006.

Profile 2. Tyson loved cars and motorcycles since he was very young; he began working on them while he was still too young to drive. His dream has always been to design cars and motorcycles. When he graduated from high school he found that his high school grades, and especially his math background, were not good enough to be admitted to a university engineering program. He worked while taking evening courses at a community college for two years before transferring into a university engineering program. Tyson found his physics courses interesting, but struggled with math. Tyson studied engineering for another four years before he graduated with an engineering degree. In school, he learned that there were very few job opportunities to actually design cars and motorcycles. However, Tyson had done a senior project using a **rapid prototyping** (RP) machine. The RP machine could automatically build almost any part that Tyson could design on a computer. He learned to create many different types of part designs on the computer using what is called computer-aided design (CAD) software. With this software Tyson could make dimensional drawings, and he spent many extra hours in the lab designing and using the RP machine to make his designs (Figure below). After graduation, he took a job in Texas with a rapid prototyping company. Soon Tyson found that the rapid prototyping technology could be used to make expensive specialty parts, and he began working with motorcycle designers in Italy and Spain. He also found a NASCAR racing team that needed custom parts and worked with their designers.



This piston assembly was designed using CAD software similar to the one Tyson used.

After eight years, Tyson decided to start his own company designing and producing high-end custom motorcycle and car parts. He now lives in California and owns two sports cars and a motorcycle. Tyson, 43, earned \$285,000 in 2006. He travels out of town and out of the country two or three times a month. His most important tools are his computer that has very good CAD software and his mobile communication system. Tyson enjoys listening to his music collection while he works.

Profile 3. Raji's childhood dream was to be a dolphin trainer. She really loved biology and chemistry classes in high school, but was undecided about her college major. A guest speaker in her high school biology class described how engineers were combining biology and technology to develop new technologies that could one day help blind people see; the speaker encouraged Raji to consider an engineering career. With her good grades, she received a college scholarship that paid for her tuition, room and board.



Bioengineers help design prosthetic limbs that allow amputees to live a more active life.

Raji earned a bioengineering degree in four years, and her favorite courses were those that included time in the bioengineering labs. In her junior year of college, because of her good grades and careful lab work, she received an invitation to work with a team of students and professors on a research project designing prosthetic limbs for amputees (Figure above). Raji found that she really liked research. After graduating with a bachelor's degree, she decided to go to graduate school for a PhD. Raji, now 28, will complete her PhD degree next year and hopes to work for a bioengineering company as a research engineer. She has also considered teaching at a university. Raji likes to ski and plans to begin scuba diving. Maybe Raji will finally get to swim with the dolphins.

Profile 4. Xaio grew up in Taiwan and studied many hours every day while in high school so he would be accepted into a regional college. He was very interested in how computers work and wanted to learn to design them, so he studied computer engineering in college. Xaio knew that he would be able to find a job when he finished school, but most of the jobs for computer engineers in his home region did not pay as well as similar jobs in other countries. In fact, some of the job opportunities in other countries paid more money in one year than Xaio's family made in ten years. However, such a high paying job would require a master's degree from a good school in another country, and that would be expensive. Xaio applied to schools in the United States and in Great Britain and was accepted to a good school in the United States, where he finished a master's degree in computer engineering in two years.



Designing one of the integrated circuits on this circuit board for an Apple iPod Sport is a project that Xaio might work on

Xaio has been working for an electronics design company in the United States for five years. Because of his knowledge of Taiwanese culture and language, and his knowledge of electronics design, the design company trained him in **microelectronics** manufacturing and testing. Now Xaio is a team leader for manufacturing some of the company's designs that are being made in Taiwan (Figure above). He travels to Taiwan about four times a year. His hobbies include tennis and ballroom dancing. Xaio made \$85,000 in 2006.

Profile 5. Glenn had many interests growing up; he played on a soccer team for several years, and played trumpet in his grade school and junior high bands. In high school, he was good at math and science, but he also enjoyed playing trumpet in the marching band and competing on the swim team. As a junior in high school, he had a very difficult time deciding what his college major should be; he liked many different things, and was not sure which he wanted to pursue. Several of his teachers suggested that he

consider engineering, and after visits to several colleges, he decided electrical engineering appealed to him. He started as a college freshman in electrical engineering. One year later, he decided that mechanical engineering was a better fit for his interests; he switched to mechanical engineering and graduated with a B+ grade average three and one-half years later.

After graduation, he was hired by a large aerospace company whose primary business is Department of Defense contracts. The company provided a one-year training period in which he rotated through several different divisions of the company and became familiar with the different product lines within the company. Now he works as a member of a large team updating engine and transmission designs for a military helicopter. He enjoys the technical challenges of his job. He plans to improve his technical expertise by starting a masters program in the next few years. He believes that this will help him move into a team leadership position.

Engineering Is Diverse and Global

Now that you have read the profiles of several different engineers and made a list of their attributes, have any of your original ideas about engineers changed? What have you discovered about engineers and engineering?

Hopefully, you have noticed that engineers are as diverse as the types of careers they pursue. They are women and men, young and old. They are **consultants**, teachers, and technical sales representatives. They work for small companies and large companies. Many start their own companies. They work in industrial plants and research labs. Some engineers work in an office; some work in production and manufacturing facilities; others spend most of their time working outdoors. And some engineers do a great deal of travel.

Engineers need a college degree, and many choose to acquire advanced specialization by pursuing a master's or PhD degree. Others choose to pursue an engineering degree because it provides them with both a solid technical background and strong critical thinking skills that support them in other fields such as law, medicine, business, and public service.

You may have also noticed that engineers can make a good income, that they often work in teams, and that those teams are composed of people from around the world. In the past ten years engineering has become a global career.

Activity

(For this exercise you need access to the Internet or a library.) Approximately 75,000 students graduated from engineering colleges in the United States following the 2005–2006 academic year. See if you can find out how many engineering graduates there were from other countries. Which countries have the most engineering graduates? Can you guess why?

Review Questions

The following questions will help you assess your understanding of the Discovering Engineering Section. There may be one, two, three or even four correct answers to each question. To demonstrate your understanding, you should find all of the correct answers.

1. Communication skills
 1. are as important for engineers as technical skills
 2. are not important or necessary for engineers
 3. will help you manage your team
 4. none of the above
2. An engineering degree
 1. limits your career choices to specialized engineering fields
 2. provides technical background for careers in many fields
 3. allows you to work in a variety of settings and around the world
 4. provides both a general technical background and a specialization
3. Engineering work is performed
 1. mainly in the United States
 2. mainly in Europe
 3. in countries around the world
 4. by teams of engineers distributed in many countries
4. Engineers
 1. have no interests outside of engineering

2. have many interests outside of engineering
3. all love nature and being outdoors
4. drive fast cars

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2.3: What Makes an Engineer?

Engineers solve problems using math, science, and technology. They also design products that are useful for humans. To become an engineer you need a degree in engineering that will provide you with a broad background in math, science, and technology, as engineers use these skills to solve problems on a daily basis. Besides the broad background, engineering students also choose a specialization in some branch of engineering. Engineers in each branch have knowledge and skills that can be applied to many fields and can contribute to solving many different types of problems. Since many engineering projects encompass multiple problems to solve, engineers in one field often work closely with specialists in other fields, including scientists, other engineers, and business leaders.

Engineering Specialization

Most engineering specializations have emerged over the past 200 years as scientific knowledge in various fields has grown. Prior to that, engineering focused primarily on the construction of roads, bridges, canals, or military structures and devices.

Activity

To better understand the breadth of engineering specializations it is time to make another list. This time you do not need to do any research. Simply make a list of all of the engineering specializations or types of engineers you can think of, and write a brief description of one of those specializations. Refer to the list of engineering societies that represent different engineering specializations at the end of this chapter. How many were you able to name? Are there others that you did not write down? Was your description of the engineering specialization similar to the description listed? You may want to spend some time reading all of the descriptions to better understand the various engineering specializations.

Activity

Now that you are familiar with some of the different engineering specializations and the major societies that represent engineering, let us see if you can match an engineering design project with an engineering specialization.

An aircraft manufacturer wants to design and manufacture the world's largest airplane. What type of engineer(s) should they hire?

From reading the description of engineering specializations at the end of the chapter, your first response might be an aerospace engineer. However, did you know that there are miles of electrical wiring and thousands of electronic devices inside of an airplane such as the Airbus A380 shown in the Figure below? Therefore, it might be a good idea to hire an engineer with some knowledge of electrical systems (perhaps an electrical engineer). We probably do not want the aircraft to break into pieces under the weight of the hundreds of people or thousands of pounds of cargo inside the aircraft, so it might be a good idea to hire structural engineers or civil engineers. Today there are thousands of different materials that can be used to manufacture products so we might want to hire engineers with specialized knowledge of materials (materials engineer). Pilots need to be able to operate the very specialized equipment that controls an airplane, so you might want to hire engineers who specialize in human-computer interaction (industrial engineer). It might also be a good idea to hire systems engineers who have specialized knowledge of how the different parts of the aircraft (mechanical, electrical, structural, materials, human-computer interaction) fit and work together.



The Airbus A380 is the largest commercial jetliner in the world. It can carry up to 850 passengers in two passenger decks in the fuselage.

Enrichment Activity (Quick)

Select one of the engineer profiles in the beginning of the chapter. Write a brief report that explains what type of engineering specialization, if any, you think the engineer has.

Enrichment Activity (Medium)

To better understand the engineering specializations, go to the websites of one or more of the professional societies and read about the specialization. Write a report that describes the engineering specialization you selected.

Engineering Skills

Many employers hire engineers because of particular skills, and not because of a particular discipline, degree, or specialization. Let us explore the range of engineering skills and educational degrees that employers look for in their employees. Job advertisements usually describe a position and list the skills, experience, and education required or desired for the position. Engineering skills can be highly technical, and may include the ability to use certain types of math and science, the ability to use certain types of instruments, the ability to operate certain types of computer programs, or the ability to apply certain areas of specialized engineering knowledge.

Activity

At the end of the chapter you will find several engineering job advertisements that were posted on the Internet in 2007. As you read them, you may notice terms that are new or unfamiliar to you, particularly if the ad is describing a specialized technical skill. You may also see terms that you do understand. Read each of the job descriptions and requirements carefully. Make a list of the degree requirements for each position, the experience required for each position, and the skill requirements that you understand. Did you notice that an engineering degree was listed as a requirement in all three ads? You might have also noticed that none of the positions required a specific engineering specialization.

About half of all engineering job advertisements today do not require a discipline-specific engineering degree. Rather they require an engineering degree coupled with a set of specific skills or experience.

Two of the ads list a desired number of years of experience, and all of the ads list specific types of experience. Below you will find one example from each of the ads.

- **Ad 1:** Experience in managing complex, high-profile projects.
- **Ad 2:** Familiarity or experience in one or more of the following areas: product development, program management, imaging and printing.
- **Ad 3:** Experience in injection molding plastics

Experience is a very important qualification for most engineering jobs. Many engineering students gain experience while they are in school through internships and/or through part-time employment. Others gain employment experience after school and progress to new positions as they gain more experience.

Let us now look at some of the engineering skills with which you are probably more familiar. Did you notice that all three ads require good communication skills?

- **Ad 1:** Demonstrates strong communication skills by clearly documenting activities and presenting information, technical content and ideas through spoken and written words; listens well.
- **Ad 2:** Good communication skills
- **Ad 3:** Strong communication skills with the ability to initiate establish and maintain positive relationships with internal and external customers. Clean, accurate, precise work and documentation.

Engineers must be able to communicate their ideas to others. Engineers often make presentations, write technical reports, and interact with customers and other technical experts.

One of the ads uses the following words: “clean, accurate, precise work and documentation.” Many engineers keep detailed notebooks of their work. This helps them remember how they solved a problem, or why they chose to design a product a certain way. Do you think the Wright Brothers kept good notes while they were trying to design the world’s first airplane? They recorded every experiment, every failure, and every success. Sometimes engineering notes are used to apply for **patents** that can be quite valuable. Sometimes engineers must defend their designs when problems occur. Why do you think it would be important to have engineering notes and documentation in the case of an engineering failure, such as the collapse of a bridge or a building? One answer is that notes and documentation help engineers find the causes of failure, which ultimately leads to improved designs. Another answer is that good documentation can protect engineers against lawsuits.

All three ads also required good organizational skills.

- **Ad 1:** Defines and prioritizes realistic, specific goals; able to complete scheduled tasks in the face of changing priorities.
- **Ad 2:** Good organizational skills, multitask ability, teamwork ability a must, self-directed
- **Ad 3:** Detail-oriented, strong organization skills, time management (time lines), and deadline driven. Self-starter, motivated, and proactive.

Engineers frequently work on multiple projects simultaneously (multitasking), and most of those projects have different tasks and corresponding deadlines. Engineers also usually work with one or more teams simultaneously, where each team member has different skills and responsibilities. Task deadlines are critical to the success of most projects. Sometimes missing a deadline can cause an entire project to be cancelled, or may result in the loss of significant revenue. For example, imagine that you are on a team designing a new video game controller. If you do not finish the design, testing and manufacture of the product, your company may miss the holiday season in which the majority of product sales will occur. Or perhaps your company knows that another firm is also designing a new video game controller and that the first company to get their product to market is likely to acquire the most customers.

Enrichment Activity (Medium)

Look at five engineering job openings on a job posting website or in the newspaper and list the specific qualifications of those five positions. Are there qualifications that they all have in common?

Enrichment Activity (Medium)

Identify one or two engineering skills from the advertisements below that interest you, and do some research to explain the nature and details of that skill.

One ad listed the following requirement:

- **Ad 1:** Uses a logical, systematic approach to solving problems through analysis and evaluation of alternate solutions.

Engineers learn to solve problems using a careful systematic problem-solving approach. Note that the requirement also states, "...and evaluation of alternative solutions." Usually, there is more than one solution to a problem.

Activity

A fire has been burning in a coal mine for several years in the northeastern United States. As shown in the two images (Figure below and below), the fire is completely underground; smoke rises through cracks in the ground in some areas and the ground has collapsed in several locations. There are many potential solutions to this problem: we could fill the mine with water; we could try to smother the fire by cutting off oxygen; or we could just let it burn.

There are many possible solutions to most problems, and in order to ensure the best solution is selected it is important that engineers evaluate each and every alternative. In the situation above, which of the solution to the mine fire do you think would cost the most? Which solution would cause the most harm to the environment or to the people that live in the area? Which solution is most likely to actually put out the fire? These are the sorts of questions engineers must answer to arrive at an optimal solution. The solution that was actually chosen for the mine fire was to let the fire continue to burn.



A fire in an underground coal mine in Centralia, Pennsylvania, has been burning since 1962.



Smoke rising up through cracks in the pavement caused by the intense heat of the fire burning below.

Engineering Education

In 2006 there were approximately 350 engineering colleges or schools in the United States and Canada. There are hundreds more in other countries. Most engineering colleges or schools have multiple engineering programs that offer degrees in different engineering specializations. For example, Arizona State University (ASU) in Tempe and Mesa, Arizona, offers the following 12 engineering and engineering technology degrees. In addition, within many of these degrees are specialized concentrations or focus areas.

- Aerospace Engineering
- Bioengineering
- Chemical Engineering
- Civil and Environmental Engineering
- Computer Engineering
- Electrical Engineering
- Electronics Engineering Technology
- Engineering (multidisciplinary)
- Industrial Engineering
- Manufacturing Engineering Technology
- Mechanical Engineering
- Mechanical Engineering Technology

Engineering programs are usually accredited by an organization outside of the university. Accreditation is like a stamp of approval, indicating that the engineering program has been evaluated, and that it meets standards for a quality process, adequate resources, and an appropriate engineering curriculum. The largest accreditation organization for engineering programs is ABET. In 2007, ABET accredited more than 2,700 different programs in engineering, technology, applied science, and computing.

ABET requires that all engineering programs demonstrate that their students attain the outcomes shown in Table 1. These outcomes are quite general, and are needed by almost any engineer. In addition to these outcomes, there are specific outcomes required by each engineering discipline. Thus, electrical engineering students must demonstrate the ability to design complex electrical and electronic systems; mechanical engineering students must demonstrate the ability to design and realize thermal and mechanical systems. Finally, each engineering program may have outcomes that are specific to the program; for example, these outcomes may address the needs of companies or industries that hire the program's graduates. If you study engineering in an ABET-accredited program, you will spend part of your time pursuing each of these different outcomes.

Student outcomes required by ABET of every engineering program.

Outcomes:

- an ability to apply knowledge of mathematics, science, and engineering;
- an ability to design and conduct experiments, as well as to analyze and interpret data;
- an ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability;

Outcomes:

- an ability to function on multidisciplinary teams;
- an ability to identify, formulate, and solve engineering problems;
- an understanding of professional and ethical responsibility;
- an ability to communicate effectively;
- the broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context;
- a recognition of the need for, and an ability to engage in life-long learning;
- a knowledge of contemporary issues;
- an ability to use the techniques, skills, and modern engineering tools necessary for engineering practice.

To prepare to study engineering in college, you should get a good foundation in high school in math and science. Look at Outcome (a) in Table 1 (an ability to apply knowledge of mathematics, science, and engineering). In most college engineering programs, students study chemistry, physics, higher math (either calculus or discrete mathematics), and possibly biology; so you need to be prepared to enter into college-level study in these areas. You will be best prepared if you take high school courses in all of these areas. In addition, it is helpful if you are ready to start calculus while in your first year in college.

Completing an engineering curriculum can be challenging and will probably require many hours of study outside of the classroom. Most undergraduate engineering programs are designed so that you can complete a bachelor's degree in four years. Because they were not well prepared in high school or do not stay on track as a full-time student, many students, however, take five or more years to complete their degrees.

Completing an engineering degree with good grades opens up many possible gratifying career paths. An engineering degree provides the foundation for the types of careers discussed in the profiles in the section "Some Practicing Engineers." Many graduates of engineering programs also move into technical sales or engineering management positions within the first ten years of their careers. In addition, an engineering degree provides an excellent starting point for graduate education. Many people with a bachelor's degree in engineering choose to pursue a master's degree in an engineering specialization to gain advanced and deep knowledge. A bachelors degree in engineering also provides a good foundation for an advanced degree in law, business, or even medicine.

Enrichment Activity (Long)

Identify two different university engineering programs, preferably at different schools. Research the admission requirements and the classes you would be expected to take in the first two years of the program. How many of these courses are math, science, and engineering courses? See if you can identify the courses in each of these areas.

Engineering Licensure

Many engineers choose to become licensed as a professional engineer (PE). While licensure is not required for the majority of engineering careers, only licensed professionals are allowed to offer their services to the public and sign and seal plans for the public. Some engineers who are not required to hold a professional license choose to do so for other reasons; for example, some do it to demonstrate that they have accomplished a recognized standard. It may also be advantageous when seeking or changing employment opportunities, as the PE certification sets a candidate apart from other nonlicensed engineers.

The requirements for engineering licensure are determined by each state, and therefore vary somewhat from state to state. Despite the variations, there is a standard process. The first step is to graduate from an ABET-accredited engineering program. The second step is to take the engineering fundamentals exam (FE) that covers the fundamental engineering sciences that are studied in engineering school. The third step is to acquire experience through employment, the criteria for which varies from state to state. The final step is to take the engineering professional examination.

Review Questions

The following questions will help you assess your understanding of the Discovering Engineering section. There may be one, two, three, or even four correct answers to each question. To demonstrate your understanding, you should find all of the correct answers.

1. Engineers often work closely with
 1. other engineers
 2. business people
 3. teachers
 4. the public
2. In the past 200 years, engineers have specialized because
 1. there were too many people in one field of engineering
 2. the space program needed more inventions
 3. knowledge of science and technology has increased
 4. they could make more money as a specialist
3. If you were an aircraft manufacturer which engineers would you need to hire?
 1. Structural
 2. Electrical
 3. Aerospace
 4. Material
4. An engineering skill is
 1. the same as an engineering specialty or discipline
 2. the ability to use certain kinds of instruments
 3. the degrees you have earned such as a master's degree
 4. the number of years you have worked at an engineering firm
5. Engineering experience can be gained by
 1. taking extra courses
 2. reading outside of class
 3. taking things apart at home
 4. internships
6. Engineers use their communication skills to
 1. apply for jobs
 2. make presentations
 3. interact with customers
 4. ask for a raise
7. Engineers keep a notebook to document their work so that they have
 1. data to gain a patent
 2. data for lawsuit
 3. information to review their designs when problems occur
 4. data to prove hours worked
8. Engineers work
 1. only in very large companies
 2. only with other engineers
 3. almost always alone
 4. in one or more teams
9. Engineering teams consist of engineers with different
 1. responsibilities
 2. skills
 3. designs
 4. deadlines
10. Engineering problems have
 1. one best solution
 2. partial solutions
 3. more than one solution

4. complex solutions
11. Engineering programs are accredited by an organization called
 1. ENGG
 2. ASU
 3. NSF
 4. ABET
12. Students in engineering programs must have
 1. an understanding of professional and ethical responsibility
 2. knowledge of contemporary issues
 3. a broad education
 4. ability to engage in lifelong learning
13. Most engineering societies
 1. help engineers find employment
 2. have a strong educational component
 3. require advanced degrees for membership
 4. are sponsored by businesses
14. A well-prepared student who studies engineering full time will take
 1. as little as two years to complete a degree
 2. five plus years to complete a degree
 3. six years to complete a degree
 4. four to five years to complete a degree
15. Calculus is important to engineering. Students who do not take calculus in high school
 1. can never become an engineer
 2. should take calculus in their first year of college
 3. can skip calculus if they have taken algebra
 4. can substitute physics for calculus
16. Courses in an engineering major may include
 1. calculus
 2. physics
 3. chemistry
 4. biology

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2.4: The Global and Societal Impact of Engineering

Engineering has had an impact on all aspects of society. Look around you and notice all of the things that have been made by humans. Through designing, manufacturing, testing, or selling, an engineer probably had something to do with most of these human-made items.

Great Engineering Achievements

Activity

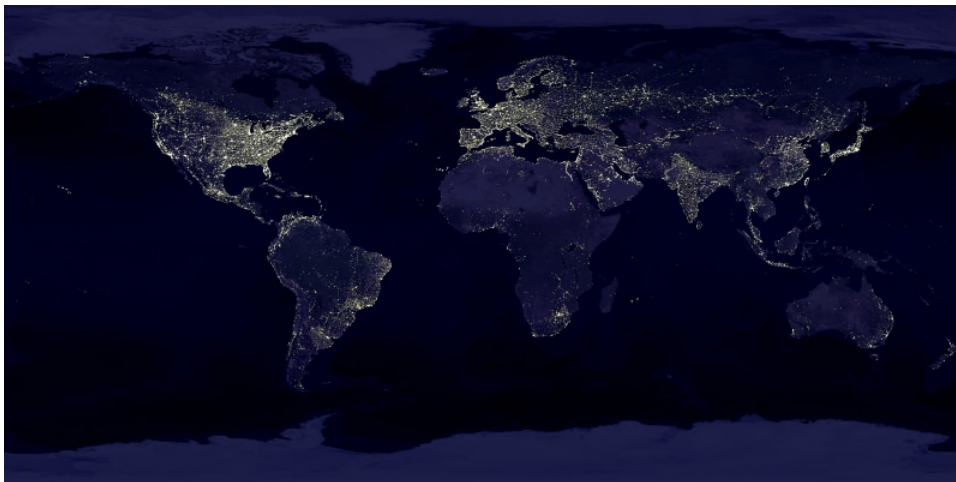
Can you think of some great engineering achievements? Take a few moments to make a list of some of the most important things engineers have developed. It might help to think of things that have changed the way that people live. For example, a century ago people relied on candles and lanterns for light. How has this changed? When you are finished making a list, share it with someone else and find out what they think are the most important engineering accomplishments.

Now that you have a list of great engineering achievements, see what others have identified as the most important accomplishments of this century. In the following, you will find several figures, each representing a significant engineering accomplishment of the twentieth century. Look at each figure carefully and try to determine what engineering accomplishment it represents. Check to see if you have the accomplishment on your list. If it is not there, add it. Each accomplishment is briefly discussed after the figure.

The National Academy of Engineering (NAE) has identified the top twenty engineering achievements of the twentieth century. The NAE has created a webpage (<http://www.greatachievements.org/>) which describes these achievements and the impacts that these achievements have in the everyday lives of people. Many of these achievements are so commonly used in our society that we take them for granted. We describe ten of the twenty achievements; the other ten achievements can be found at the NAE webpage.

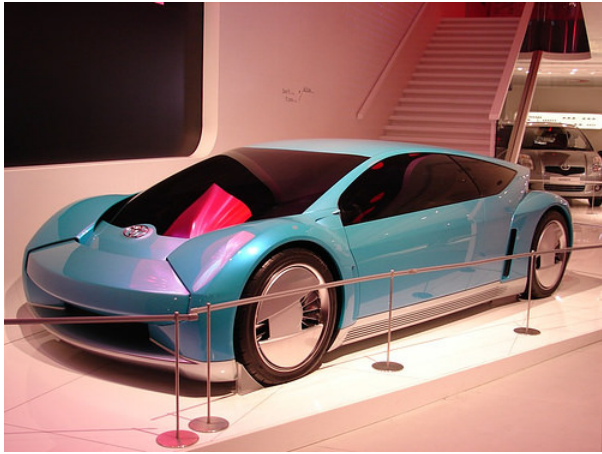


The skyline of the Pudong New Area in Shanghai, China at night.



The lights of major cities around the world are visible from space at night.

The Figure above shows the bright lights of the Pudong New Area in Shanghai, China. The next Figure above shows lights visible from space at night represent electrification. Electrification is the process of making electricity available to large numbers of people. We use electricity not only for light, but also to power machinery, instruments, and appliances. How many electric or battery powered devices do you use in a day? Without electrification, we would not have any of these devices today.



A Toyota concept car.

The Figure above shows a rather high-tech looking automobile. The first cars produced in the United States were sold in 1901, primarily as novelties to the wealthy. However, by 1920 automobiles were mass-produced. Prior to the automobile people worked close to where they lived; one had to live in the city in order to work in the city, as the largest distance that it was practical to travel regularly was only a few miles. A farm or a factory that was not close to a city could not easily transport goods to market. Thus the automobile is credited with freeing people from the limitations of geography and with greatly contributing to raising incomes and wealth.



A high flying jet aircraft leaves contrails in the sky. A contrail is the white streak (or cloud) formed behind a high-flying aircraft's engines.

Figure above shows a jet aircraft and its contrails as it flies high in the sky. Airplanes further freed people from the constraints of geography by making rapid long-distance travel possible. Airplanes are also responsible for advancing a global economy.



Clean drinking water flowing from a faucet.

Figure above shows a water faucet, and represents the supply and distribution of clean water. Clean water has had a significant impact on human life. During the 1700s and 1800s, thousands of people died from diseases including cholera, typhus, and waterborne typhoid fever, and thousands upon thousands became ill. A clean water supply and good distribution not only improved health, but also contributed to the growth of new cities, the development of hydropower, the improvement of crop growth, and the availability of water recreation.



An electronics workbench.

Figure above shows an electronics workshop. Our world is filled with electronic devices, including computers, mobile phones, music players, cameras, calculators, ATMs, and televisions to name a few. We use electronics for communication, entertainment, manufacturing, to diagnose disease, to help us drive our cars, and for thousands of everyday activities.



A television that was manufactured in 1953.

The image (Figure above) shows an early television, manufactured in 1953. Radio and television are electronic devices that deserve special attention because of their impact on the way news and information are communicated. Prior to the development of these technologies, news and information traveled slowly, through written forms of communication. Today, the television allows people to view world events in real time. With its hundreds of channels, people can also experience other lands and cultures and be entertained.



An irrigation system waters growing cotton plants.

The image (Figure above) shows an irrigation system for a large farm, and represents agricultural mechanization (the development of machines that help farmers produce crops). Prior to the development of farm equipment, farmers relied on animals to help them plow their fields. The planting, watering, and harvesting of crops was all done by hand. The amount of work required to produce crops limited the crops that individual farmers could grow. This also meant that many people were employed in farming and many families grew their own produce. Machines made it possible for a single farmer to produce larger quantities of crops, as well as a more consistent quality of crops. This, in turn, provided greater supplies of food for society, and reduced its cost.



An HB85B computer, manufactured in the early 1980s.

The image (Figure above) shows an early computer. Computers change the way we communicate. Computers help us write; this chapter has been written, formatted, and distributed by computer. In engineering and science they perform complex computations; there are many problems, such as weather prediction, that require billions of computations. Without computers, we could not do these complex calculations. Computers are also used to control machines. Computers help guide and fly airplanes; they control the engine in your car. Computers can store vast amounts of information that is readily available, and they connect us to the world through the Internet. Computers facilitate learning, and provide us with a great source of entertainment.



The LifeStraw is a water purification device designed to filter bacteria out of water and is powered by suction. Water is passed through an iodine-coated bead chamber that kills bacteria and parasites. It costs around \$3.75 and can last for a total of 700 liters of water.

The image (Figure above) shows a woman drinking water through a filtration straw, and represents an example of healthcare technology. The specialized straw is capable of filtering harmful bacteria and parasites from polluted water supplies. In the past decades, there have been numerous healthcare technologies developed that have decreased **mortality rates**, increased life spans, and contributed to a better quality of life. These technologies include advanced surgical techniques, artificial organs, instruments that can diagnose ailments, and preventive healthcare devices.



Walls of apartment buildings.

The image above (Figure above) may be the most difficult to discern. The picture shows the side of a large building with air-conditioning units on many windows. Air-conditioning was originally developed to help cool manufacturing processes. In the mid-1900s, home air-conditioning was developed, fueling an explosive growth in Sunbelt cities such as Las Vegas, Houston, and Phoenix. Air-conditioning has changed our work environments, permitting us to work in greater comfort. It has also shifted the patterns of seasonal work and play.

The ten other great engineering accomplishments of the twentieth century identified by the NAE include highways, spacecraft, the Internet, imaging, household appliances, health technologies, petroleum and petrochemical technologies, laser and fiber optics, nuclear technologies, and high-performance materials.

Enrichment Activity (Medium)

Write a brief report about one of the great engineering achievements of the twentieth century from the list earlier in this chapter. Give some specific examples of how the achievement has changed the way that people live and explain why the achievement is important.



Before the telegraph, telephone and automobile, messages were sent by horseback. This figure shows the official seal of the Post Office Department, the predecessor of the United States Postal Service.

The Impact of Engineering

To understand the impact of engineering on society we can imagine how people lived 100 years ago before these technologies existed. For example, how did people communicate without telephones and the Internet? The primary method of long-distance communication was letters. While letters are a wonderful means of communication, they take time to write and even more time to be delivered. If the distance between sender and recipient was great, it may have taken months to deliver a letter via Pony Express (Figure above).

Advancements in communication have also helped change the way many companies work today. Remember the profile of Ashley, our first engineer? Ashley works at home managing two engineering teams from across the world. That would not have been possible with letters. Engineering solutions have continually improved the quality of life, added business value, and significantly influenced the **global economy**.

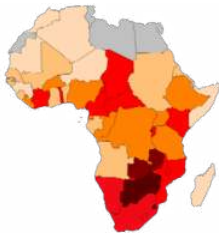
Engineering has both intended and unintended consequences. For example, air-conditioning makes comfortable life possible in much of southern United States. However, sometimes the unintended consequences of new technologies can be negative. About a decade ago, scientists discovered that Freon and similar gasses used in air conditioners were contributing to damage to the Earth's protective ozone layer. As a result, new gasses and technology had to be developed. Consider as well the impact on culture from air conditioners. Prior to air-conditioning, many people sat on their front porch in the evenings, in part because their homes were too hot. Also, people often had very high ceilings—a design intended to help with home cooling.

Another example of unintended consequences is several years ago a company developed corn seeds that were highly resistant to weed killers and insects so that farmers would not need to spray poisons on their fields. An unintended consequence was that the new type of corn seed, after it had been growing for several years, started growing in fields where it had not been planted. Farmers tried to kill the unwanted corn plants but were unable to do it because the corn was resistant to the poisons.

The Future of Engineering

It is very difficult to predict the future of engineering, but engineers attempt this whenever they design new products. Engineers try to determine what people will want and need—both now and in the future—and then they design things to fulfill those wants and needs.

While we do not know exactly what will happen in the future, we can examine some possible scenarios. Consider natural catastrophes. There have been many significant catastrophes in the past decade including powerful hurricanes, earthquakes, and tsunamis that have killed hundreds of thousands of people and destroyed a great deal of property. If we go back further in the history of the world we also find that major volcanic eruptions and rare collisions with meteorites have impacted the entire planet. Engineers are working on ways to protect people and property from these disasters as well as ways to predict and respond rapidly to these types of disasters.



Percentage of adults (ages 15-49) in Africa infected with the HIV/AIDS virus in 1999. The countries highlighted in light colors had less than 2% infected, while the countries highlighted in dark colors had over 20% and up to 30% infected. Grey countries had no data available.

Another threat to people's well being comes from disease. Between 1300 and 1500, the bubonic plague killed between one-third and one-half of Europe's population. Later, cholera killed large numbers as well. Today, these diseases have been largely eradicated in the developed world through engineering of clean water and sanitation systems. However, the world currently faces an AIDS epidemic (Figure above), and there are likely to be new disease threats in the future. Engineers will work with scientists, governments, and health workers to develop and implement technologies that will prevent and respond to these threats.

Another certain need that will be met by engineering is energy. Because all people in the world need energy, the world production and use of energy is growing at a rapid pace. You might recall that both electrification and the automobile were listed as great engineering achievements of the twentieth century. Fossil fuels, the source of energy used most often to produce electricity and to power automobiles, also causes pollution and contributes to global warming. Also, supplies of fossil fuels are limited and becoming more expensive. Some wonder if the world can sustain the current energy growth and consumption patterns. With only a few countries owning the majority of energy resources, there is further concern about the supply of energy at prices that most people can afford. Engineers and scientists are working on developing new energy technologies for the future.

Some emerging trends in engineering are in the areas of biotechnology and nanotechnology. In the area of biotechnology, engineers are working on designs that impact the human body, animals, and plant life. Engineers and scientist are working on technologies to help the blind to see, the hearing impaired to hear, and the disabled to walk. Biotechnology has also opened the possibility of controversial areas such as cloning. Engineers are also working with scientist to develop crops and processes that can be used as fuels for energy.

Nanotechnology refers to the development of products and components that are very small, typically between 1 and 100 nanometers. A nanometer is 1×10^{-9} meters. A nanometer is so small that it takes a very powerful microscope to see an object of that size. While the area of nanotechnology is very new, it shows promise to provide new technologies ranging from lighter and stronger materials to nanorobots that can repair individual cells to new treatments for cancer.

Enrichment Activity (Long)

Envision the future by designing and drawing a picture of a new product. Explain what need or purpose the new design is fulfilling. Ask others to review your design and give you some feedback. Then use the feedback to redesign your product.

Review Questions

The following questions will help you assess your understanding of the Discovering Engineering Section. There may be one, two, three or even four correct answers to each question. To demonstrate your understanding, you should find all of the correct answers.

1. An emerging area of engineering is
 1. nanotechnology
 2. geotechnology
 3. aerotechnology
 4. retrotechnology
2. Electrification is
 1. generating and distributing electricity to many different users
 2. being shocked by electricity
 3. powering machinery, instruments, and appliances with electricity
 4. not an important engineering accomplishment
3. The automobile and jet airliner
 1. have freed people from the limitations of geography
 2. are both important engineering accomplishments
 3. have the same type of engines
 4. are dangerous and should be abandoned
4. Computers
 1. provide us with entertainment
 2. control machines
 3. perform complex computations
 4. store vast amounts of information
5. Air-conditioning
 1. has not had any unanticipated negative consequences
 2. contributed to damage of the Earth's ozone layer
 3. was originally developed for use in automobiles
 4. makes life comfortable in the southern United States
6. The future of engineering is
 1. easy to predict
 2. determined by engineers trying to design things to fulfill peoples wants and needs
 3. tied to energy production
 4. not affected by natural disasters

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2.5: Nature of Engineering Conclusion

In this chapter we have explored the nature of engineering through the profiles of several practicing engineers, through engineering job advertisements, and through some of the great accomplishments of engineers. We have explored some of the very broad sets of engineering careers and tasks. We have also discussed the most common engineering specializations of today. These activities establish a foundation for understanding the impact of engineering on our society. We have also examined several trends to help us understand the future of engineering. And finally, we have explored the educational requirements and licensure of engineering.

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2.6: Nature of Engineering Vocabulary

Consultant

A person who gives professional or expert advice.

Electronics

The development and application of devices and systems involving electricity.

Global economy

The international spread of trade and commerce across national boundaries with minimal restrictions from governments.

Microelectronics

The design, development, and construction of electronic systems from extremely small components.

Mortality rate

A measure of the number of deaths in a population during a given time period in general or due to a specific cause.

Patent

The exclusive rights granted by a government to an inventor to manufacture, use, or sell an invention for a certain number of years.

Rapid prototyping (RP)

The automated construction of physical objects under computer control using specialized equipment.

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2.8: Student Supplemental Resources

Engineering Job Opportunities from the Internet August 2007

Each of the following engineering job opportunities was posted on the Internet in August 2007.

Job Description 1

Engineer, Manufacturing

SPECIFIC DUTIES:

- Develops, implements and maintains methods, operation sequence and processes in the manufacture or fabrication of parts, components, subassemblies and final assemblies.
- Interfaces with design engineering in coordinating the release of new products.
- Estimates manufacturing cost, determines time standards and makes recommendations for tooling and process requirements of new or existing product lines.
- Maintains records and reporting systems for coordination of manufacturing operations.

OTHER REQUIREMENTS:

- Uses a logical, systematic approach to solving problems through analysis and evaluation of alternate solutions.
- Acts in a professional manner even when dealing with time demands and/or interpersonal conflict.
- Maintains a flexible approach toward task scheduling.
- Demonstrates strong communication skills by clearly documenting activities and presenting information, technical content and ideas through spoken and written words; listens well.
- Follows through on tasks to completion.
- Defines and prioritizes realistic, specific goals; able to complete scheduled tasks in the face of changing priorities.

QUALIFICATIONS:

- BS in Engineering plus related experience; two years for Eng; four years for Sr. Eng.
- Sound knowledge of manufacturing techniques and process control.
- Familiarity with the process of integrating technical designs into a production environment.
- LEAN Six Sigma implementation experience.
- Experience in managing complex, high profile projects.
- Knowledge of SAP applications is a plus.

Job Description 2

Media R&D Engineer (Electronics Industry)

JOB DESCRIPTION

- Technical position specializing in large format printing & LF Media development and testing. This person will be responsible for R&D deliverables during the New Product development process and also be in charge of the Large Format Image Permanence Activity.
- New Product Development is driven by cross-functional teams and requires the ability to work cooperatively and coordinate activities across the organization. This position will interact often with marketing, sales, manufacturing, Printer development teams, strategic manufacturing sites, and customers, as well as other internal personnel in conducting product and application training.
- Max. 25% travel.

REQUIREMENTS/QUALIFICATIONS

Musts:

- A self-motivated, success oriented, and the will to deliver against goals.
- Strong team interaction ability
- Strong technical skills
- Familiarity or Experience in one or more of the following areas:
 - Product Development

- Program Management
- Imaging and Printing
- Image Permanence (Fade and Durability Testing)
- Imaging materials manufacturing and testing
- BS degree in an Engineering discipline.
- Large Format printing experience - large format printing, graphic production processes, applications testing and materials selection.
- Demonstrated test development, specification establishment and troubleshooting experience.
- Intermediate computer/digital workflow skills
- Good communication skills
- Good organizational skills
- Multi-task ability
- Self-directed
- High energy
- Teamwork ability a must

Strongly Desired:

- Ability to operate large format printers is desired. Ability to develop application relevant test methods and make recommendations for improvements is a plus.
- Demonstrated ability to work well across cultural boundaries as well as multiple geographical sites and time-zones.

Job Description 3

Eyewear Test and Development Engineer

DESCRIPTION

- Interface with Designers, Research Engineers, Manufacturing Engineers and Quality in testing raw materials, prototypes and pre-production samples for manufacturing process implementation and production hand-off.
- Define, track and execute testing deliverables to ensure timely test results based on product timelines.
- Support the field-testing program which provides detailed, meaningful feedback from real-world testing.
- Be able to identify and troubleshoot processes of experimentation, redesign experiments accordingly, and design valid experiments to ensure 'killer' data.
- Identify and solve problems found through materials, product and process testing.
- Research on materials and material specifications, process technology, concepts, and new testing equipment and methods.
- Support the research, testing and implementation of manufacturing processes and parameters with appropriate work instructions and documentation.
- Identify critical process variables and parameters to develop and design experiments accordingly.
- Work in R&D lab on designing and/or implementing tests on mechanical and physical properties as well as performance properties on all related raw materials. Raw materials include: polymers, coatings and metal alloys.
- Document all testing, results, and other information pertinent to product development.
- Responsible for Design of Experiment (DOE)
- Collect data.
- Set-up test methods.
- Take ownership and manage project from beginning to end under the direction of the R&D manager.
- Responsible for full product testing.

REQUIREMENTS

- Degree in technical field (i.e. mechanical engineering, physics, chemistry, polymer/materials science) and 3–5 years of related work experience OR 6–10 years related work experience with formal technical training/certification.
- Strong communication skills with the ability to initiate, establish, and maintain positive relationships with internal and external customers.
- Experience in researching, developing and manufacturing plastics, coatings, and metal alloys.
- Experience in injection molding plastics.
- Experience with forming, casting, injecting, coating and testing metal alloys.

- Experience in mechanical and physical testing procedures to apply results to final decision making.
- Experience in executing field-testing during product development for ‘real world’ feedback.
- Requires minimal direction in executing test, experiments, test parameters, and collection of data and research.
- Detail oriented, strong organization skills, time management (timelines), and deadline driven.
- Clean, accurate, precise work and documentation.
- Self-starter, motivated and proactive.
- Natural inclination to think outside the box.
- Knowledge of statistics, mathematics, physics, and critical thinking skills.

Major Engineering Societies

Major Engineering Societies

Society	Society Information
Aerospace Engineering: http://www.aiaa.org/ American Institute of Aeronautics and Astronautics (AIAA)	AIAA is a leading advocate for aerospace among government decision-makers—and a trusted information resource for the media on all subjects relating to aerospace technology. Since 1972, AIAA has contributed extensive technical expertise and policy guidance to Congress and the Executive Branch. We testify before the House and Senate on the full spectrum of aerospace issues.
Agricultural Engineering: http://www.asabe.org/ American Society of Agricultural and Biological Engineers (ASABE)	ASABE is an educational and scientific organization dedicated to the advancement of engineering applicable to agricultural, food, and biological systems. Agricultural, food, and biological engineers develop efficient and environmentally sensitive methods of producing food, fiber, timber, and renewable energy sources for an ever-increasing world population.
Architectural Engineering: http://content.ae-institute.org/inside/intro.html Architectural Engineering Institute (AEI, part of ASCE)	AEI is the home for all professionals in the building industry. They provide a multidisciplinary national forum for members of but not limited to the architectural engineering, structural, mechanical, electrical, and architectural communities. Recognizing the necessity for a place to examine issues and exchange views and information with one another, AEI works to facilitate the crucial communication among members of the building team, both on a technical basis and in the professional arena.
Automotive Engineering: http://www.sae.org/servlets/index Society of Automotive Engineers (SAE)	SAE was founded for the purpose of advancing mobility on land, sea, air, and space. Many years ago, SAE noticed that graduating engineers were well versed in textbook knowledge and engineering theory. Surprisingly, however, college engineering curricula provided no ways for students to gain practical experience with manufacturing and production of their designs. Since this type of experience is vital for success on the job, SAE began to organize and sponsor competitions which emphasize a hands-on approach to the engineering.
Biomedical Engineering: www.bmes.org/ Biomedical Engineering Society (BMES)	In response to a manifest need to provide a society that gave equal status to representatives of both biomedical and engineering interests, BMES was incorporated in Illinois on February 1, 1968. The purpose of the Society is to promote the increase of biomedical engineering knowledge and its utilization.

Society	Society Information
Chemical Engineering: http://www.aiche.org/ American Institute of Chemical Engineers (AIChE)	AIChE is a nonprofit professional association that provides leadership in advancing the chemical engineering profession. Through its many programs and services, AIChE helps its members access and apply the latest and most accurate technical information; offers concise, targeted award-winning technical publications; conducts annual conferences to promote information sharing and the advancement of the field; provides opportunities for its members to gain leadership experience and network with their peers in industry, academia, and government; and offers members attractive and affordable insurance programs.
Civil Engineering: www.asce.org/asce.cfm American Society of Civil Engineers (ASCE)	Today's civil engineer uses every advantage to meet the demands of their profession. That is why ASCE pioneers new programs, policies, educational activities, and professional resources to help them successfully compete in their business. That is why today's civil engineer has a home at ASCE.
Computer Engineering: http://www.computer.org/portal/site/ieeecs/index.jsp IEEE Computer Society	The IEEE Computer Society's vision is to be the leading provider of technical information, community services, and personalized services to the world's computing professionals. The Society is dedicated to advancing the theory, practice, and application of computer and information processing technology.
Electrical Engineering: http://www.ieee.org/portal/site Institute of Electrical and Electronics Engineers (IEEE)	Through its global membership, the IEEE is a leading authority on areas ranging from aerospace systems, computers and telecommunications to biomedical engineering, electric power, and consumer electronics among others. Members rely on the IEEE as a source of technical and professional information, resources, and services. To foster an interest in the engineering profession, the IEEE also serves student members in colleges and universities around the world.
Environmental Engineering: http://www.aeee.net/ American Academy of Environmental Engineers (AAEE)	AAEE is dedicated to excellence in the practice of environmental engineering to ensure the public health, safety, and welfare to enable humans to coexist in harmony with nature.
Geological Engineering: rock.geosociety.org/egd/index.html Geological Society of American, Engineering Geology Division	The Engineering Geology Division promotes education, research, outreach, and application of engineering geologic knowledge toward betterment of human society by adopting sound design of buildings, structures, and facilities that assure public safety and a healthy environment.
Industrial Engineering: http://www.iienet2.org/Default.aspx Institute of Industrial Engineers (IIE)	IIE is the world's largest professional society dedicated solely to the support of the industrial engineering profession and individuals involved with improving quality and productivity. IIE is an international, nonprofit association that provides leadership for the application, education, training, research, and development of industrial engineering. IIE's primary mission is to meet the ever-changing needs of industrial engineers, which includes undergraduate and graduate students, engineering practitioners and consultants in all industries, engineering managers, and engineers in education, research, and government.

Society	Society Information
Marine Engineering: http://www.sname.org/ Society of Naval Architects and Marine Engineers (SNAME)	SNAME is an internationally recognized nonprofit, technical, professional society of individual members serving the maritime and offshore industries and their suppliers. SNAME is dedicated to advancing the art, science, and practice of naval architecture, shipbuilding, and marine engineering, encouraging the exchange and recording of information, sponsoring applied research, offering career guidance and supporting education, and enhancing the professional status and integrity of its membership.
Mechanical Engineering: http://www.asme.org/ American Society of Mechanical Engineers (ASME)	Today's ASME promotes the art, science and practice of mechanical and multidisciplinary engineering and allied sciences around the globe. ASME codes and standards strive to be the world leader in mechanical and multidisciplinary engineering codes, standards, conformity assessment programs, and related products and services.
Mining Engineering: www.aime.org/ American Institute of Mining, Metallurgical, and Petroleum Engineers (AIME)	The goal of AIME today is to advance the knowledge of engineering and the arts and sciences involved in the production and use of minerals, metals, materials, and energy resources, while disseminating significant developments in these areas of technology.
Nuclear Engineering: http://www.ans.org/ American Nuclear Society (ANS)	ANS is a not-for-profit, international, scientific and educational organization. The core purpose of ANS is to promote the awareness and understanding of the application of nuclear science and technology.
Petroleum Engineering: http://www.spe.org Society of Petroleum Engineers (SPE)	The mission of SPE is to collect, disseminate, and exchange technical knowledge concerning the exploration, development and production of oil and gas resources, and related technologies for the public benefit; and to provide opportunities for professionals to enhance their technical and professional competence. The vision is to be a society of professional excellence, providing its members the highest quality lifelong learning, and continuous personal and professional growth.
Systems Engineering: http://www.incose.org/ International Council on Systems Engineering (INCOSSE)	INCOSSE is a not-for-profit membership organization.. Its mission is to advance the state-of-the-art and practice of systems engineering in industry, academia, and government by promoting interdisciplinary, scalable approaches to produce technologically appropriate solutions that meet societal needs.

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CHAPTER OVERVIEW

3: Engineering and Society

- [3.1: About Engineering and Society](#)
- [3.2: To Engineer is Human](#)
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- [3.4: Water and Engineering](#)
- [3.5: Engineering and Society Vocabulary](#)
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3.1: About Engineering and Society

A characteristic of human beings is that we develop tools and technology to adapt our environment to meet our needs. Engineers contribute to the development of many innovations that improve life.

In this chapter, we will begin by investigating how engineers work to meet human needs; we will describe a few of the greatest engineering accomplishments of the past and consider needs that engineering must meet in the future. We then consider an example of how engineering has dramatically improved health and well-being for most of the developed world: providing clean drinking water as well as treatment for wastewater.

Chapter Learning Objectives

After working through this chapter, you should be able to

- explain the relationship between engineering and societal needs,
- explain how engineering has influenced water systems in our lives,
- describe the impact of an engineering solution related to water from an ethical viewpoint.

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3.2: To Engineer is Human



The ancient pyramids are an engineering marvel. The great pyramid was the tallest man-made structure in the world for almost four thousand years.

Engineering is a human endeavor. Humans have engaged in engineering to meet their needs as long as they have had needs. We invent and innovate when we are confronted with problems, needs, and desires. Inventions exist and continue to be created to meet our needs for daily life, such as access to water, energy, transportation, and entertainment. Petroski (2004) said, “Making things is an activity as old as civilization, and making ever new things is part of being human.” Our lives are permeated with technological inventions that humans have engineered (Figure above). Yet, the fundamental nature of engineering or what engineers do is not everyday knowledge. For instance, many of us are unfamiliar with the basic principles used to build the systems that deliver water, gas, or electricity to our homes.

Engineering is the design, analysis, and creation of things that are practical and useful in our lives. At its core, engineering incorporates design. Design, simply put, is creating something that has not existed before. Engineers help design, create or change almost everything we encounter in our lives, including what we feel, eat, see, and hear.

In this section, we consider several different engineering achievements that have dramatically changed the way we live today. We then consider the impact of engineering on sustainability of natural resources, and focused briefly on one very important resource: water.

Human Needs Are Met By Engineering

To stay alive, humans and animals need only the basics of water, food, and shelter. In addition to these basics, animals, and at times humans, too, need protection from other animals that are predators. For the most part, however, humans today do not have to worry about becoming food for other animals. Although we need water, food, clothes, and shelter, not very many of us fetch water from a well or a river; grow or kill our own food; make our own fabric and sew clothes; or, build our own homes with our own hands. We do not do this on our own because people have invented products, tools and systems to do it for us, allowing us to spend productive time in other ways.

We prefer to do something other than figure out how to get clean water to our homes, grow crops, raise poultry, make fabric and sew clothes, or construct homes. This is possible because of all the people before us who created inventions that met their and our needs. These inventions range from the basic technology of the plow or the wheel to today’s advanced technologies that allow us to manage and sustain our natural resources or engage in space exploration. Human needs and wants have been met throughout time by engineering achievements that have significantly transformed and impacted our lives, especially over the past one hundred years.

The National Academy of Engineering (NAE) has identified the top twenty engineering achievements of the twentieth century; these achievements are documented in the book by Constable and Somerville (2003) and on the webpage (<http://www.greatachievements.org/>). These achievements are described in more detail in the chapter Nature of Engineering. These achievements include many of the technologies that we use every day and may take for granted.

One of these achievements that we will consider in more detail in this chapter is water supply and distribution. Water is vital for our lives. When we turn on our kitchen faucet and clean drinking water flows from the faucet, we seldom pause to think how that water

got there (Figure below). However, delivering clean water to your home or school or office requires an advanced system of treatment and distribution facilities. Imagine your life without indoor plumbing. Yet, even in today's world, there are many who live without access to running water, toilets, bathtubs, or showers. Maintaining water quality and distributing water to people where they live are timeless challenges. It is important to remember that water is not merely needed for our basic survival; in today's world water is used for many other purposes that sustain our way of life.



Most people in the United States are fortunate to have safe drinking water.

Engineering Sustainability for Our Future

Current projections predict that the earth's human population will be around 9 billion in 2050. By the year 3000 the world's population is predicted to be double that of today's 6.6 billion. Most of this expected increase is predicted to be concentrated in the developing nations of Africa, Asia, and Latin America. Meeting the most basic needs of this future population means an ever-growing stress on our environment and limited natural resources. This means increased production and consumption of goods and services and increased demand for land, energy, and materials (NRC, 1999). While the engineering achievements of the past century have increased the standard of living in developed countries, the challenge of the future is to alleviate poverty and raise the standard of living for all on this planet while also sustaining our natural resources.

As we consider the engineering achievements of the past, we also need to consider their unintended consequences on our lives and our planet. We must ask ourselves: What are the social, economic, and environmental impacts of our engineering achievements? Many engineering decisions cannot be made without consideration of nearby natural and man-made systems, because contemporary engineering systems can affect the environment far into the future. There is a strong need to reduce the risk and level of unwanted disturbances to natural resources and our man-made world associated with engineering systems.

According to the National Resource Council (NRC, 1999): "Sustainable development—the reconciliation of society's developmental goals with its environmental limits over the long term—is the most recent conceptual focus linking the collective aspirations of the world's peoples for peace, freedom, improved living conditions, and a healthy environment." Engineers have an obligation to meet the basic needs of all humans for water, nutrition, energy, sanitation, and health, as well as the protection of the planet's resources, including our cultural and natural diversity.

Water



Water covers 71% of the Earth's surface. This photograph of Earth's polar ice and oceans was taken by the Galileo space probe.

The next section focuses on one of the basic human needs: water. Water is a common substance. Its chemical formula of H_2O is widely known. It is a life-giving natural resource. Seventy-one percent of the earth's surface is covered in water, most of which is salt water (Figure above). Water is essential to most organisms on Earth. For instance, 50–70% of a human body is made up of water.

Water is one of the few substances on the planet Earth that can be found occurring naturally in all three states: solid (ice), liquid (water), and gas (water vapor). A significant difference between these states is the density of each state. (Density here refers to how close the water molecules are to each other. The mass of water within a specific volume determines the density of water.) The vapor state is least dense, while the liquid state is most dense; the solid-state (ice) is less dense than the liquid state. Warm water is less dense than cold water. Therefore, heating and cooling water affects its density.

Water is a very good solvent for many compounds. Therefore, almost no pure water exists in nature; most water contains dissolved substances. After water falls to earth in the form of **precipitation**, it dissolves substances from soil and rocks. This leads, for example, to hard water, which is water that has magnesium and calcium dissolved in it. Next time you buy bottled water, read the label. You may discover, to your surprise, that your pure spring water is not merely H_2O , but also contains chemicals. In fact, the label on your bottled water may even list the chemicals and their amounts. Most water you drink has ions in it. Those ions in the water make it acidic or basic. The pH of water is a measure of its acidity or alkalinity; The pH index is related to the amount of free hydrogen ions in the water.

Water is a natural resource and sustaining it for future populations is a major challenge that will require creative engineering. Water is used by industry, agriculture, and homes. In many places in North America, people have access to all of the clean water they need. However, there are many places in the world where water supply is limited. Most uses of water actually decrease the quality of the water; this is true of water that is used in many industrial processes as well as water used by humans.

Our family, our community, our culture, our geographical area, and the prevailing economy often influence our values regarding water. For instance, if you are from an agricultural area, where your livelihood depends on the availability of water for your agricultural needs, you may rate the value or significance of water for your own life higher than someone whose livelihood is built around an entertainment industry such as running a cinema multiplex.

Activity

Take a few minutes to consider the various uses for water you have encountered in your life. Get out a pen and paper and list as many uses for water as you can. Compare your list to someone else's list. How are they different? What do they have in common?

Review Questions

The following questions will help you assess your understanding of this section. There may be one, two, three, or even four correct answers to each question. To demonstrate your understanding, you should find all of the correct answers.

1. Technology refers to
 1. computers, software, and computer peripherals
 2. changing the natural world to satisfy needs
 3. tools used by engineers to design systems
 4. pre-engineering courses taken in high school
2. When looking at large-scale problems engineers should focus on
 1. the scientific aspects
 2. ethical aspects
 3. economic aspects
 4. legal aspects
3. Engineering
 1. began as a twentieth-century invention
 2. has always been a human activity
 3. is a field first developed in Europe
 4. began with Alexander Graham Bell
4. Engineers help design things that are
 1. practical and useful in our lives
 2. only useful to scientists
 3. too specialized to be useful
 4. not practical or economical
5. One of the biggest challenges of engineering is to
 1. raise the standard of living while using natural resources
 2. keep the standard of living the same while sustaining natural resources
 3. keep the standard of living the same while using natural resources
 4. raise the standard of living while sustaining natural resources
6. Engineering decisions must take into consideration
 1. internal systems
 2. natural systems
 3. man-made systems
 4. future systems
7. Sustainable development is linked to
 1. basic needs
 2. freedom
 3. nutrition
 4. world peace
8. Which of these influences our values regarding water
 1. culture
 2. cost
 3. cleanliness
 4. family
9. Engineering achievements can have
 1. mistaken consequences
 2. unrealistic consequences
 3. no consequences
 4. unintended consequences

10. When we use water we
1. decrease its quality
 2. increase its quality
 3. do not affect its quality
 4. improve its quality

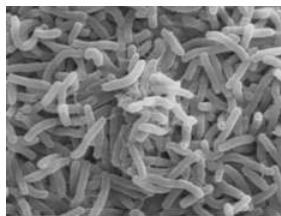
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3.3: Water and Disease- A Case Study

Do you ever wonder how diseases spread? Water serves as a carrier or transmitter of many diseases. In fact, many of the world's diseases are dependent on water for their transmission. These diseases often occur in large **epidemics** and can be quite catastrophic. A large part of water engineering is maintaining water quality so the water that we receive when we turn on our kitchen faucets is **potable**. In addition to making water safe for drinking, water engineering is also about transporting water from its source into our homes and to the areas where we live.

In this section we will investigate cholera, a waterborne disease; we will learn about the scientific problem solving that verified that cholera was transmitted by water. We will then begin to investigate how engineers help society manage our water supply. This topic will be expanded in later sections.

Background Information: Cholera



Microscopic view of cholera bacteria.

Cholera is an acute intestinal infection caused by ingestion of contaminated water or food. The infection is caused by the *Vibrio cholerae* bacterium (Figure above). This bacterium typically travels through water that is contaminated by human or animal feces. Human beings can contract cholera by ingesting water or food contaminated by the *V. cholerae* bacterium. Symptoms of cholera include diarrhea, abdominal cramps, nausea, vomiting, and dehydration. As the human body does not produce lasting immunity against the cholera bacterium, it is possible that the disease can be contracted more than once. When people die from cholera, it is generally due to severe dehydration caused by the illness. When left untreated cholera generally has a high mortality rate. A cholera outbreak can be widespread, causing a large scale epidemic in which hundreds or thousands of people die.

Activity

Research and review resources on cholera epidemics on the World Wide Web. Refer to encyclopedias from your school library or local library. Prepare a short list of answers that do not exceed 200–300 words for each of the following questions. You might use some of these suggested phrases for searching on the Internet: cholera, epidemics, cholera and water, water-borne diseases, water contamination sources, water quality, water purification.

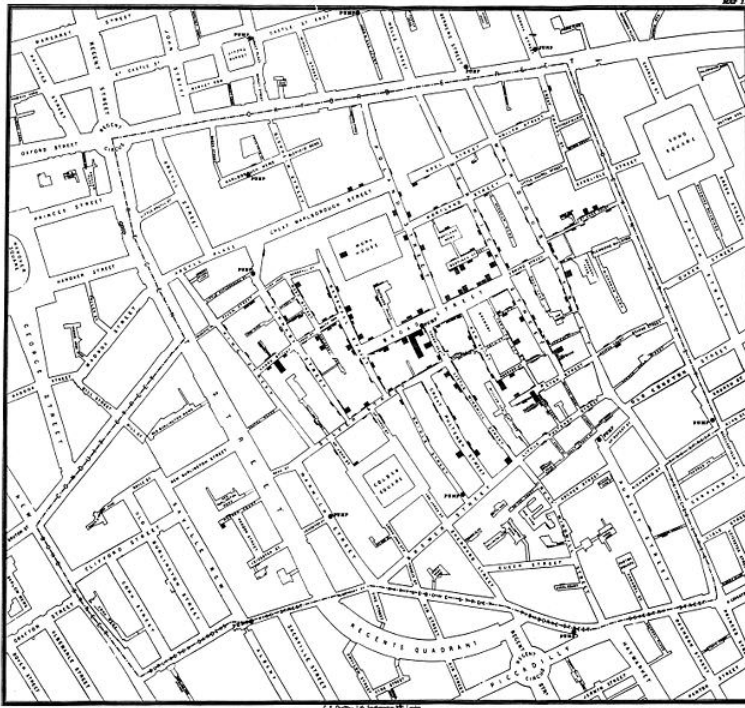
- What were some of the major cholera epidemics of the past?
- What are some of the major cholera epidemics of the present?
- Where and under what conditions did the cholera epidemics occur?
- What role did water play in these cholera epidemics?
- How can the quality of water be maintained so the water does not get contaminated from bacteria such as the cholera bacterium?

The 1854 Case of Cholera in London

In 1854, hundreds of people living in London died during a cholera epidemic. At that time, very little was known about how the disease was transmitted. Many believed that people could contract diseases such as cholera by breathing bad or foul air. It is estimated that 616 people died during this epidemic. If not for the intervention of Dr. John Snow (b. 1813–d. 1858), who carefully mapped the number of deaths that occurred around water sources, primarily the Broad Street pump, and then worked with other officials to remove its handle, the cholera epidemic of 1854 in London would have been more widespread. It was later discovered that sewage was contaminating the water source for the Broad Street pump. Preventing people from using this pump stopped the use of water from that source. This action has been credited with greatly reducing the number of deaths during this epidemic.

Dr. Snow created the map in the Figure below. The Snow map depicted the area around the Broad Street pump where the number of deaths is indicated by dark blocks at the homes and establishments around the pump.

During an earlier cholera epidemic in London in 1848 and 1849, Snow had proposed the unusual idea that cholera was caused by something that was ingested orally, rather than by inhaling foul air. During the cholera epidemic of 1854 in London, Snow collected data by carefully interviewing people around Broad Street who had not been infected by the water they had been drinking. Others would cite, after Snow's death in 1858, that the data he had collected supporting his theory that cholera was primarily spread by sewage-contaminated water. Dr. Snow's research into the spread of cholera locally around the Broad Street pump is an important lesson taught even today to epidemiology students as a model of scientific reasoning. Dr. Snow authored a report about the cholera outbreak in the Parish of St. James, Westminster, during the Autumn of 1854; this as well as his other works are available online at the John Snow Archive and Research Companion website.

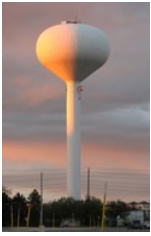


Number of deaths at 40 Broad Street. Deaths at homes are indicated by dark lines.

Today most of us know that unclean water carries bacteria that cause disease. We value clean water. However, in the 1800s, the idea that water could carry disease was frightening to many. During that time many people believed that catastrophic diseases such as cholera were visitation of retribution for sinful living of the poor. Dr. Snow had to work hard to convince the local authorities to remove the handle of the Broad Street pump. Now, people know that cholera infection can be avoided by drinking clean water. During the twentieth century, in the western hemisphere, incidences of cholera in the recent past are almost nonexistent. However, it is estimated by the World Health Organization (WHO) that in developing nations only 35% of the population have access to clean water.

According to the WHO, the current response to cholera outbreaks tends to mostly in the form of emergency response whenever there is an outbreak. While emergency response is very important and prevents many deaths, it does not help prevent causes of cholera. Therefore, one cannot emphasize enough the importance of long-term cholera prevention. Key to the prevention of cholera and other waterborne diseases is closely related to the prevention of water contamination and water purification. Separating sewage water from natural water sources, treatment of sewage water, treatment of water resources, and purification of water for storage and delivery for human consumption are key elements of such a strategy that involve engineering solutions. In addition, medical research, personal hygiene, and public health education will play a key role in the prevention of cholera epidemics.

Water Engineering and Management



Water towers such as this one help distribute clean water to surrounding communities.

One of our significant challenges today is to obtain water for people and for food production, look after critical ecosystems and habitats, and deal with the variability and uncertainty of water on our earth. Water engineers and managers need to understand the complexity of water use and water resources management. Water for human consumption includes engineering solutions related to water supply and sanitation; management of water treatment and distribution systems (Figure above); storm water, domestic wastewater, and urban drainage; and sustainable management of water for urban and rural areas. Water for food production includes the use of water for agricultural purposes which necessitates engineering solutions related to planning, design, implementation, operation, and maintenance of irrigation and drainage projects. Water for energy production includes the use of engineering solutions such as dams for hydroelectric power. Water management is also related to engineering solutions for uncertain events such as flooding caused by natural and human activities.

Review Questions

The following questions will help you assess your understanding of this section. There may be one, two, three, or even four correct answers to each question. To demonstrate your understanding, you should find all of the correct answers.

1. Water engineers are concerned with
 1. ensuring that water is drinkable
 2. drinking eight glasses of water a day
 3. transporting water to our homes
 4. marketing bottled water
2. An example of a disease caused by contaminated water is
 1. a cold
 2. colessus
 3. Cholera
 4. colitis
3. We can prevent waterborne diseases by
 1. washing our hands
 2. public health education
 3. vaccinations
 4. visits to the doctor
4. Water engineers look for solutions to mitigate
 1. overfishing the oceans
 2. natural flooding
 3. building ships
 4. human caused flooding

Reflection Questions

- Why do North Americans not have to worry about cholera?
- When we travel abroad to some countries, we are advised to receive a cholera vaccination. Why is it necessary to receive a cholera vaccination?

Further Reading

- Johnson, Steven. *The Ghost Map (The Story of London's Most Terrifying Epidemic and How It Changed Science, Cities, and the Modern World)*. Riverhead Books, New York, 2006.

- Tuthill, Kathleen. “John Snow and the Broad Street Pump: On the Trail of an Epidemic.” Cricket, 31 (2003): 23–31. Available on the web at
- <http://www.ph.ucla.edu/epi/snow/snowcricketarticle.html>

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3.4: Water and Engineering

Many of the great engineering achievements beginning at the earliest times through the present have been associated with providing a reliable water supply and with disposing of wastewater. The Roman aqueducts that supplied Rome and other cities with drinking water between 300BC and AD 300 are one example of this. The sewer system in London developed during the late 1800s is another example. Because of these engineering accomplishments, most people in the developed world have access to safe drinking water.

In this section, we investigate the relationship between engineering and water. We first describe the concept of water quality standards and how drinking water should meet these standards. We then describe the techniques engineers have developed to provide potable water and to treat wastewater so that it does not unduly **pollute** the environment.

Water Quality Standards

Most of us know that the water we receive from our local municipalities is treated. The treatment of water has to follow water quality standards. In the United States, the development of water quality standards began in the early twentieth century. Over time, these water quality standards have evolved and have become more rigorous. Laws and regulations such as the Clean Water Act govern water quality standards in the United States. The US Environmental Protection Agency (EPA) has the mission of protecting human health and the environment. The EPA has created water quality standards that define the goals for a **body of water** and specify its uses. They establish criteria to protect those water uses. They also establish requirements to protect water from pollutants.

In general, a water quality standard consists of four basic elements (EPA):

1. Designated uses of the water body (e.g., recreation, water supply, aquatic life, agriculture).
2. Water quality criteria to protect designated uses (numeric pollutant concentrations and narrative requirements).
3. An antidegradation policy to maintain and protect existing uses and high-quality waters.
4. General policies addressing implementation issues (e.g., low flows, variances, mixing zones).

State governments are required to identify the appropriate uses for water bodies. These uses are identified by taking into consideration the value of the water body for various uses. The uses for a water body can include any of the following: public water supply, protection of fish, shellfish, and wildlife, recreation, agriculture, industry, and navigation. In making these designations, the appropriate authorities must examine whether the water body is suitable for the intended use. This determination is made based on the physical, chemical, and biological characteristics of the water body, its geographical setting and scenic qualities, and economic considerations.

Drinking Water



The city of New York monitors its drinking water to ensure that it is safe. This sign marks the location of a station where the drinking water is sampled.

The quality of drinking water differs from one location to another. Quality is dependent on the condition of the water source and the treatment. Each local community water supplier is required to provide an annual report, sometimes known as a “consumer confidence report” to the public. This report usually provides information on the quality of the local drinking water, its source, and the **contaminants** found in the water (Figure above).

Scientists report the contaminants in water as contaminants in parts per million (ppm), parts per billion (ppb), and parts per trillion. When water contaminant measurements are reported in this manner, you may wonder: How can amounts of contaminants that seem so very small cause health problems? Scientific research has shown that, for example, lead concentrations as small as 15 ppb can be

harmful to infants and children. The smell of petroleum products in water can be detected by humans even when the amount of petroleum in water is as low as 10 ppb.

It is very important to note that there is no such thing as naturally pure water. All water in nature contains some sort of impurities. Since water is a good solvent, as it flows through various layers of soil and over rocks, it dissolves minerals. These minerals in the water may get the taste (good or bad). However, if these minerals are above a certain level, they could be harmful.

Have you ever thought about where your drinking water comes from? Many of us take the drinking water we get when we turn on our kitchen faucet for granted. The water in many urban areas comes from surface sources such as lakes, rivers, and reservoirs. These sources can be near the urban areas or far away. Water suppliers procure and treat the water they provide. It is important to consider the entire watershed that provides the water. A watershed is the land from which precipitation (usually rain or snow) flows into the river, lake, or reservoir.

In many urban and rural areas, water also comes from ground water that is pumped from a well. Wells provide water from aquifers, which are natural water reservoirs under the earth's surface. Aquifers may lie under parts of several states or may only be a few miles wide. Thus, water quality is affected not only by the water source such as a river or well that we can see, but also by activities that occur many miles away.

Before the water is delivered to our homes, the water is treated. The treatment of drinking water that is most common is disinfection. Water suppliers add chlorine or another disinfectant to kill bacteria and germs in the water. This commonplace drinking water treatment is considered to be one of the most important scientific advancements of the twentieth century.

Activity

Common water treatment methods: Before we examine a water treatment plant, list and describe some common ways to treat water or purify water for drinking that you can do at home or during emergencies.

Materials needed: Internet-accessible computer, science textbooks.

Suggested phrases for searching on the Internet: Water purification at home, Water purification during emergencies, Common water purification methods.

Directions: Working alone or in a group, brainstorm a list of ways in which drinking water might be bad. From your list, identify problems that might need to be treated. Brainstorm a second list of ways to obtain good drinking water in your home. Do any of these solve the problems identified on your first list?

Using the Internet, research news stories related to water supply and water problems in your local community. How do the results of your research compare to your first list of potential problems with water? Are there problems that you did not anticipate?

Research information on water purification techniques that can be used at home. Discuss in small groups what you learned from your research. Prepare a chart with a brief description of common water problems and the treatment/purification methods appropriate to each problem; include a list of materials you would need for each treatment method and illustrate each method by making your own drawing.

Hints: Filtration, Boiling, Distillation, Water Softeners, Reverse Osmosis

Extensions: What will be the cost of implementing these home-based water purification techniques if you were to need clean water for a family of four? How would you go about determining the amount of water that you will need? What is the most effective method in terms of removing impurities for the quantity of water you need for a family of four for drinking/food purposes?

Water Treatment Basics

Municipalities must perform two types of water treatment: treatment of water before it is used for drinking and other purposes, and treatment of wastewater such as sewage so that it is safe to return to the environment. Typically, drinking water is treated based on the quality of the source water. Ground water, for instance, usually requires less treatment than surface water from a river or a lake. Thus, the quality of the water that enters your community water treatment plant determines how the water will be treated prior to distribution.

Wastewater includes sewage as well as water used in industrial and agricultural processes. It flows out of homes and neighborhoods through sewage pipes to a wastewater treatment plant. Here, the wastewater is treated to remove solids and contaminants. The treated wastewater may be re-used for irrigation and landscaping. The treated wastewater is also returned to streams, rivers, and oceans, which can also be a source of pollution.

Treatment plants for drinking water and for wastewater have equipment and processes to remove or destroy harmful materials and organisms. A treatment plant uses tanks and mechanical parts such as valves and pumps to move the water through the different processes. These processes are designed and managed by water engineers. Other engineering specialties, including mechanical and construction engineering, play a role in the construction and maintenance of water treatment plants.

Drinking Water Treatment

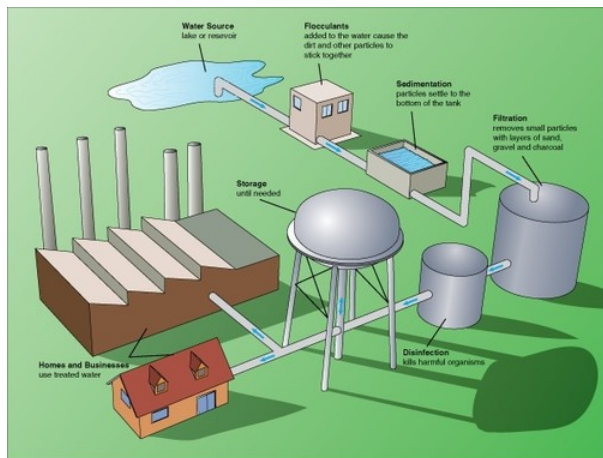


Illustration of a typical drinking water treatment process.

Water from rivers, lakes, and streams or ground water is pumped and transported to a drinking water treatment facility. Then the water is processed through various units to prepare the water for distribution to homes. The water treatment involves the steps shown in Figure above and described below.

Coagulation. First, dirt and other particles must be removed from the water. Flocculants are chemicals such as alum (aluminum potassium sulfate) that cause the dirt and other particles to stick together; flocculants are added to the water, which creates larger particles called floc.

Sedimentation. As the water moves through the sedimentation tanks, the floc particles settle to the bottom of the tank. The clear water then flows to a filtration unit.

Filtration. Filtration removes small particles from the water by passing it through layers of sand, gravel and charcoal. The water then moves to disinfection before storage.

Disinfection. Water is disinfected with chlorine or other chemicals, called disinfectants, to kill any bacteria and other harmful organisms. The amount of disinfectants added to the water has to be carefully adjusted, because too much may be harmful to humans, but too little will not kill the harmful organisms.

Storage. After disinfection, the water is stored in storage tanks until it is needed for distribution to homes, businesses, and other water users.

Activity

What are the mechanisms that maintain water pressure during peak water usage times in urban areas? You can easily imagine that in most urban cities, water usage will peak during certain times of the day. Have you wondered how the water flow rate and the pressure at which the water flows out of your home's faucets is maintained during the mornings when most people are preparing to leave for school or work? This is a time when water use is high as people brush their teeth, shower, and use water for personal hygiene.

Materials needed: Internet-accessible computer, science textbooks.

Suggested phrases for searching on the Internet: Water pressure, water flow during peak water usage times

Directions: Research and review information on how water delivery systems or local municipalities or your city's water supply systems regulate water flow and water pressure during peak times. Discuss in small groups what you learned from your research. Prepare a brief essay that describes the mechanisms or techniques used by water suppliers to maintain water flow and pressure during peak times. Illustrate this technique by making your own drawing.

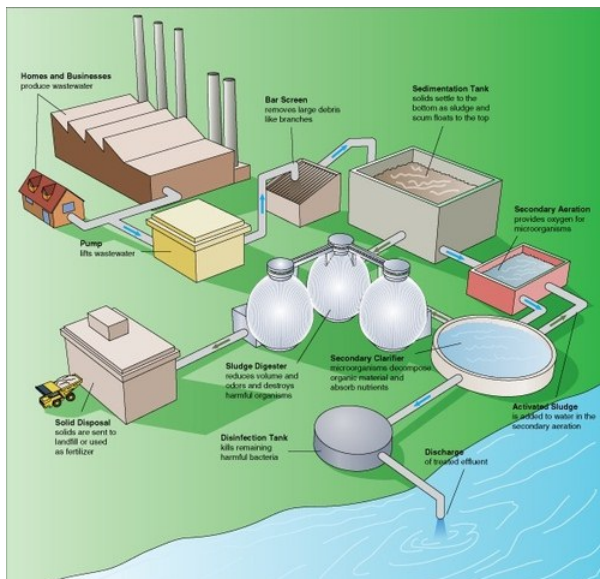
Hint: Have you seen water tower tanks in cities that are in flat areas? Research what purpose these water tower tanks serve. What do water pumps do? Research the various ways in which a city or municipal water supply system uses water pumps.

Wastewater Treatment

Before wastewater can be released into the environment, it is treated in a wastewater treatment plant. An aerial view of a wastewater treatment plant is shown in Figure below. The next image (Figure below) illustrates the steps involved in the treatment of wastewater. We describe the steps in more detail below.



An aerial view of a wastewater treatment plant in Dresden, Germany.



Steps in a typical wastewater treatment process.

Pumping. Wastewater treatment facilities are usually located on low ground so that **gravity** will move sewage from homes to the treatment plant. Usually, pumps are needed to lift the sewage as it enters the treatment facility. The treatment facility uses gravity to move the wastewater through the treatment process.

Bar screen. As it enters the treatment plant, wastewater may contain large items such as plastic bottles, cans, sticks, rocks, and even dead animals. These items are removed by the bar screen and sent to a landfill. If they are not removed, they will damage equipment in the treatment plant.

Grit chamber. After screening, wastewater enters the grit chamber in which larger particles (such as sand or dirt) settle out of the water. Often, the water is aerated (air is bubbled through it) to keep smaller particles from settling out. Aeration causes some of the gases that are dissolved in the water (e.g. hydrogen sulfide that smells like rotten eggs) to be released.

Sedimentation tank. In the sedimentation tank (also known as the primary clarifier), solids settle to the bottom as sludge and scum floats to the top. The sludge is pumped out of the primary clarifier and sent to the solids processing facility. The scum is composed of lighter materials such as grease, oil, soap, and so forth. Slow-moving rakes are used to collect the scum from the surface of the wastewater.

Secondary aeration and clarifier. The wastewater is exposed to air in an aerator, which provides oxygen for microorganisms that help break down contaminants in the water. This may be done by spraying the wastewater into the air or by bubbling air through the wastewater. The aerated effluent is passed into a secondary clarifier, which is a large tank or pond; in the clarifier, microorganisms decompose organic material and absorb nutrients such as nitrogen and phosphorus. The microorganisms and remaining solids settle out of the effluent as activated sludge. Most of the activated sludge is pumped to the solids processing facility, while the remaining sludge is pumped into the wastewater entering the aerator. This introduces additional microorganisms to the wastewater to hasten the breakdown of organic matter.

Filtration. Filtration may be used to further reduce the organic matter in the water. The water is filtered through a substance, usually sand and rocks. During this filtration process, most bacteria are removed, turbidity and color in the wastewater are reduced, odors are removed, the amount of iron content in the wastewater is reduced, and any other solids that may have remained in the water are also removed. This water may subsequently be filtered again through a carbon filter such as charcoal to remove organic particles.

Disinfection. To kill remaining harmful bacteria and other **pathogens** in the processed wastewater, chlorine and other chemicals are added in a disinfection tank. The chlorine can be harmful if added in excess quantities. (You may have noticed the smell of chlorine or have had irritated eyes when you were exposed to chlorine in a swimming pool.) Therefore, in some cases, the chlorine must be neutralized with other chemicals after it has killed the bacteria to protect marine organisms.

Solids processing. Solids include the sludge and scum removed in the sedimentation tank and the activated sludge removed from the secondary clarifier. These solids may be processed further in devices called digesters, which are heated and enclosed tanks. The solid wastes are kept in these tanks for 20–30 days to reduce the volume of the material, reduce odors, and also destroy any organisms that have the potential to cause disease. Depending on the source and composition of the wastewater, the digested solids are either sent to a landfill or used as fertilizer for crops. The use of the processed solid wastes as fertilizers is usually done only after careful testing for any potential dangerous contamination.

The treated water that is released by the plant is called effluent. The effluent is usually released into a local river or the ocean. In some places, this water may be used for landscaping (e.g. to water lawns or golf courses), but not for drinking purposes.

Review Questions

The following questions will help you assess your understanding of this section. There may be one, two, three, or even four correct answers to each question. To demonstrate your understanding, you should find all of the correct answers.

1. The United States government agency that is responsible for water quality standards is
 1. the Environmental Protection Agency
 2. the national air and space administration
 3. the president of the United States
 4. the forest service
2. Urban water supplies come from
 1. fire hydrants
 2. surface sources such as lakes, rivers, and reservoirs
 3. the ocean
 4. groundwater
3. Two types of water treatment are
 1. treatment of wastewater
 2. freezing and melting
 3. adding a slice of lemon
 4. treatment of drinking water
4. The design and management of processes in a water treatment plant involved these engineering specializations
 1. water engineering

2. social engineering
 3. mechanical engineering
 4. construction engineering
5. The purpose of disinfection is to
1. further pollute water
 2. kill fish and other marine animals
 3. kill bacteria
 4. make water taste good
6. The purpose of wastewater treatment is to
1. make the water clean enough to drink
 2. remove solid and organic matter from water
 3. kill harmful bacteria
 4. reduce pollution caused by wastewater

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3.5: Engineering and Society Vocabulary

Body of water

A significant amount of water either naturally (such as lakes, rivers, and oceans) or man-made (such as ponds, lakes, and harbors).

Contaminant

A substance that may be harmful to humans or other forms of life when released into the environment.

Epidemic

a widespread outbreak of a contagious disease.

Gravity

The force of attraction exerted between objects. Often, this is the force of attraction that the earth exerts on objects at its surface.

Pathogen

A disease-producing organism.

pH (potential of Hydrogen)

A measure of the activity of hydrogen ions (H^+) in a solution. This is a measure of the solution's acidity or alkalinity. pH is the logarithm of the reciprocal of the effective hydrogen-ion concentration and is a number between 0 and 14; the number has no units. A pH of 7 indicates neutrality, that is, the solution is neither acidic nor alkaline. pH numbers lower than 7 indicate acidity, while pH numbers higher than 7 indicate alkalinity. Each change of 1 pH represents a tenfold change in acidity or alkalinity.

Pollute

To make something dirty, foul, or unclean.

Potable

Potable water is water that is clean enough to drink.

Precipitation

Water that falls from the atmosphere to the earth's surface. The most common form of precipitation is rain. Snow, sleet, hail, and freezing rain are also forms of precipitation.

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3.6: Engineering and Society References

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CHAPTER OVERVIEW

4: Introduction to Engineering Design

- [4.1: About the Introduction to Engineering Design](#)
- [4.2: The Design Process](#)
- [4.3: The Design Process in Action](#)
- [4.4: Introduction to Engineering Design Vocabulary](#)
- [4.5: Introduction to Engineering Design References](#)

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4.1: About the Introduction to Engineering Design

Before beginning this chapter, look around you for a moment and notice how much of your environment has been designed by humans. Perhaps you are reading inside a building; people designed, engineered, and constructed it. This book has been designed; if you are reading it in a paper format, the paper, ink, and binding have been manufactured, while if you are reading it electronically, the computer on which you are reading has been designed and fabricated.

Almost every aspect of modern life depends on and is affected by technological **artifacts** such as bridges, buildings, vehicles, cell phones, computers, and so on. These technological artifacts are designed and created by engineers; the process by which engineers create is often called the engineering design process. ABET, the organization that accredits undergraduate university and college engineering programs, has developed the following definition of the engineering design process.

Engineering design is the process of devising a system, component, or process to meet desired needs. It is a decision-making process (often **iterative**), in which the basic sciences, mathematics, and the engineering sciences are applied to convert resources optimally to meet these stated needs.

Chapter Learning Objectives

After working through this chapter, you should be able to do the following:

- Explain what a design process is.
- Explain how engineering design differs from other design processes.
- Explain the different steps in the engineering design process.
- Apply each step of the engineering design process to design a product or process.
- Describe how the implementation the design process affects the quality of the resulting design.

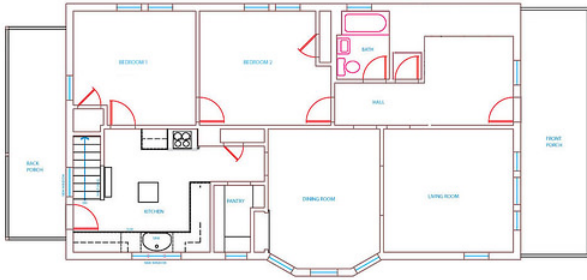
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4.2: The Design Process

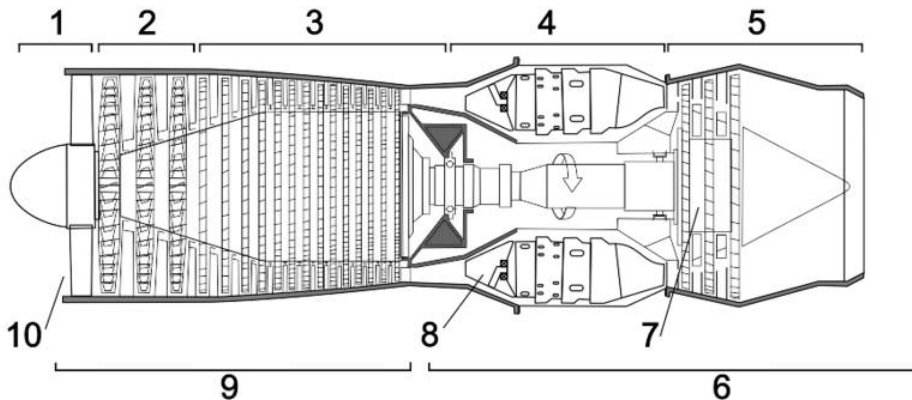
The word *design* has several meanings. One meaning is a plan or drawing that shows the look and function of an object or structure before it is made. This is the meaning of design that we consider in this chapter. Examples of this meaning for design include

- a team of architects may develop the design of a building, specifying the building's appearance and the layout of rooms, doors, and hallways;
- a team of engineers may develop the design for a new jet engine, specifying its structure and the materials from which it is built.

These images (Figure below and Figure below) show examples of this type of design.

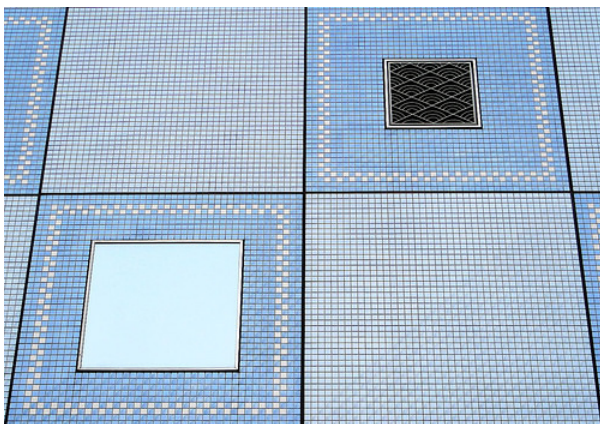


The floor plan of the first story of a house. Architects document their designs using such floor plans.



A design of a jet engine.

In general conversation, the word design is often used to mean “an arrangement of lines or shapes to form a pattern or decoration.” The image (Figure below) shows an example of this meaning for design; this is *not* the meaning that we are considering in this chapter.



A design can be an arrangement of shapes and lines that forms a decorative pattern. This is *not* the type of design that we will consider in this chapter.

A process by which a design for an object or a structure is created is called a *design process*. All design processes have similarities. They all involve creativity. They all involve making decisions. However, engineering design tends to require a more extensive and specialized knowledge of technology, math, and science than other types of design.

Engineering Design

Teams undertake several different types of engineering design projects. Most projects involve modifying or enhancing an existing product or process; these projects are described as **incremental design**. Sometimes a new product is developed from scratch. Such a project would be a new design. Both types of projects use the design process.

The purpose of most design projects is to develop an object, structure, or process that meets the needs of **customers** and **stakeholders**. A customer is someone who will use the designed object; customers usually pay for the product. A stakeholder is someone who has an interest in the product. Customers are stakeholders; other stakeholders may include government agencies, companies, and individuals. For example, stakeholders for a new automobile design might include the individuals who will purchase it, the mechanics who will maintain and repair it, the Environmental Protection Agency that will monitor its emissions, and the oil companies that will supply its fuel.

Design problems are almost always open ended—they rarely have a single correct solution. Instead, there are usually several solutions that will satisfy the desired needs. One of the challenges of design is to choose from the vast number of possible solutions. Indeed, engineered products may be very complex, and the design of such products may require that a design team make hundreds or thousands of decisions. Without a clearly defined process to follow, the team may not develop designs that meet the needs of customers or other stakeholders; the team may make poor decisions or lose sight of the important attributes of the product. Thus, it is important to use a structured design process. Structured design processes offer several advantages. They provide a framework in which the team decision-making is made explicit and the decision process can be well documented. They also reduce the likelihood that important issues will be forgotten or overlooked.

In this section, we describe “the engineering design process,” which may give the impression that there is only one correct process. In reality, many processes have been successfully applied to engineering design. The design process discussed in this section may not always be the best for a given project or problem. The design process should not be applied blindly, but should be adapted to fit the circumstances of the design team and the particular project. The design process should also be subjected to a continuous improvement process so that the design team’s performance improves over time.

The engineering design process may not result in a **viable** design for several reasons:

- Incorrect or unrealistic assumptions.
- A lack of understanding of the desired needs or underlying problems to be solved.
- Errors in design specifications or representations (e.g. **models** and drawings).
- Inadequate testing of **prototypes**.
- Poor design choices.

Sometimes, poor design decisions made early in the process will make it impossible to develop a design that successfully meets customers' needs.

A Design Process

In this section we present a typical engineering design process. Other descriptions of the design process may break the process into somewhat different steps and may use somewhat different terms to describe the steps, but most design processes are similar to this one. An example of how this design process could be applied is in The Design Process in Action section.



A design process.

Our basic engineering design process includes the steps shown in the Figure above. Note that image shows each step being completed before the next step begins; this process is sequential. In many real-life situations, the design team may revisit a step several times to create the final design. We now describe the steps in more detail.

Define the problem. In the problem definition step, the needs of potential customers are investigated; potential competitors are identified and their market positions are **characterized**; **constraints** imposed by government regulations or technological limitations are identified; and constraints on the design effort such as available personnel, time, and money are established. The problem definition process results in a clear understanding of the scope of the design project and the resources available to solve the design problem. This understanding is often expressed in a **problem statement**. This understanding is also expressed in the form of criteria and constraints.

Identify criteria and constraints. **Criteria** and constraints are used to evaluate the quality of a design. Constraints describe conditions that must be met by the design and design process; a design must meet all constraints. Criteria are measurable values that can be used to compare several designs and determine which is better.

Generate ideas. Once criteria and constraints are identified, the design team generates ideas for designs. These ideas come from many different sources; these include existing products (including competitors' products), brainstorming and other creative activities, and market and technical research. Ideas are combined to generate potential designs; at this stage, designs are concepts without a significant level of detail.

Explore possibilities. After potential designs are generated, they are explored to understand their characteristics and likely advantages and disadvantages.

Select a design concept. Potential designs are evaluated relative to the constraints and criteria, and one or more is selected to be designed in detail and prototyped. This selection is made using a structured process that requires the constraints to be met and chooses the best design according to the criteria.

Develop a detailed design. The selected design is developed in more detail. The **design architecture** is established by identifying physical and functional chunks. Shapes and **dimensions** are determined, materials and fabrication processes are selected, and product components are identified. The design is developed in enough detail that prototypes and models of the design can be made.

Create models and prototypes. One or more prototypes are typically implemented to characterize various aspects of the design. Prototypes may be physical models of the design in which dimensions, materials, and fabrication processes emulate important aspects of the design. Increasingly, prototypes are implemented using computer modeling software that simulates mechanical, electrical, and other characteristics of the product.

Test and evaluate. Prototypes are tested to see whether the design meets all constraints and performs acceptably relative to the criteria.

Refine the design. Testing and evaluation may reveal weaknesses of the design or indicate ways in which the design may be improved. At this point, the design may be refined to better meet the criteria and constraints. Sometimes, testing and evaluation show that a design will not work, so that a different design concept must be selected; in this case, the process goes back to the "Select a design concept" step.

Implementation. Depending on the context, the design is produced or constructed.

Communicate process and results. The activities and results of the design process are documented. This documentation is communicated to the appropriate stakeholders in the design.

The design process in the Figure above is often called a sequential process because each step follows the previous one in direct sequence. This model does not account for the iterative nature of many actual design projects; as designs are developed, prototyped, and evaluated, their strengths and weaknesses are better understood and changes are made to the design on the basis of this improved understanding. After changing the design, the process of prototyping and evaluating is repeated. The design process in which the processes are iterated is often called a spiral design process. A spiral design process is illustrated in the Figure below.



A spiral design process. Changes are made in the design, and then the improved design is evaluated.

The advantage of using a spiral design process is that the end design is often much better than the initial design. The significant disadvantage of the spiral design process is that time and resources are required for each loop in the spiral; if these are not planned for, the project may easily be late and over budget.

Review Questions

Multiple Choice

The following questions will help you assess your understanding of the Discovering Engineering section. There may be one, two, three, or even four correct answers to each question. To demonstrate your understanding, you should find all of the correct answers.

1. Which attributes describe the engineering design process?
 1. Creativity
 2. Specialized knowledge of math and science
 3. Decision making
 4. Specialized knowledge of technology
2. Incremental design means that the
 1. design process is slow
 2. an existing product or process is modified
 3. design is done in pieces
 4. an existing product or process is enhanced
3. The design process may not produce a good product because
 1. the team was made up of different kinds of engineers
 2. materials were delivered too late to use
 3. the engineers focused on attributes of the product
 4. prototypes were not tested adequately
4. Which of these is used to evaluate the quality of a design?
 1. The number of prototypes tested
 2. The spiral design process
 3. Criteria and constraints
 4. The size of the engineering firm
5. What is the first step in the design process?
 1. Explore possibilities
 2. Define the problem
 3. Select a design
 4. Generate ideas
6. What is the last step in the design process?
 1. Communicate process and results
 2. Test and evaluate
 3. Refine the design
 4. Implement
7. A good design
 1. does not have constraints

2. is not limited by criteria
3. meets constraints
4. develops criteria and constraints

Free Response Questions

8. Why are planning and evaluation as important as creativity in the design process?
9. What is the difference between engineering design and other types of design (architectural, fashion, etc.)?

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4.3: The Design Process in Action

In this section, we go through an example of a team using the design process. This section provides more detail about the steps of the sequential design process.

Case Study

To provide concrete examples throughout this section, we will use a design case study. In this case study, we will follow the design process used by an intrepid team of engineers who work at a small manufacturing company to develop a product that solves some of the problems with current commuting options. At the beginning of their project, the team chose a suitable engineering name for their project: the **sustainable** commuter vehicle, or SCV for short.

According to the United States Department of Transportation, in 2000, over three-fourths of the trips made to and from work were made by individuals traveling alone in a car, sport utility vehicle, or truck. In some ways, jumping in the car and going is the hallmark of modern American life. Americans prize the convenience and comfort of the modern automobile, even though it creates some serious problems.

- Vehicles create 20% of **greenhouse gas** emissions in the United States and appear to be a significant contributor to future climate change. **Carbon dioxide** is one of the principle greenhouse gasses emitted by vehicles.
- A typical American household spends more money on driving costs than it spends on food.
- In most major metropolitan areas, “rush hour” (congested traffic conditions similar to those in the Figure below) now lasts six to seven hours a day.
- Traffic congestion costs \$63.1 billion per year. Each year, commuters stuck in traffic jams waste 2.3 billion gallons of fuel, not to mention their time or frustration.



Rush hour traffic in Washington, D.C. Heavy traffic and long delays, as well as the associated air pollution and fuel consumption, are major problems for communities.

Is there a way to solve some of these problems without completely giving up the comfort and convenience that we have come to expect?

Activity

To increase understanding of the issues faced by the engineering team, complete one or more of the following exercises that involve problems associated with commuting.

- *Research the issues associated with commuting in your area. These issues might include traffic congestion, accidents, and pollution.*

- Do you know anybody who commutes regularly to work alone in their automobile? Talk to them about the benefits and drawbacks associated with this.
- Find information on transportation planning in your area. How much money is spent developing new roadways? How much money is spent upgrading and maintaining existing roads? Is the current road network effective?
- What problems do environmental activists in your area see with the current commuting **infrastructure**?

With your understanding, write a paragraph describing the commuter problem from the perspectives of the commuter, the city planner, and the environmental activist.

Activity

Spend an hour working with a team of classmates to develop a design solution to the commuter problem. Write a paragraph that describes what your team did during the hour. Then consider the following questions:

- Did your team find a solution? If not, why not?
- What processes did your team use to find solutions?
- How good is your solution? How do you know whether it is good or not?
- How well did you document your design process?

In this section, we will describe the (fictional) design process used by the SCV team to address the commuting problem.

Define the Problem

Problem definition is one of the most critical steps in the design process. Since the design team trying to solve this problem will expend a significant effort, it is very important that the problem being addressed is actually the problem that is important to potential customers. It is also important that the problem be clearly defined and understood by the design team.

Many techniques can be used to clearly define and understand the problem (see Fogler and LeBlanc, 1995). These techniques include

- gathering information from customers and other stakeholders,
- finding expert information (either in person or through books or other sources),
- doing a **root cause analysis** to identify what the real problem is.

The SCV design team began by gathering information about the issues associated with vehicular commuting and traffic congestion. They found and read several government reports. They interviewed various stakeholders in the commuting problem; these included people who commute to and from work in their car each day, officials from state and local departments of transportation, and representatives of environmental groups. They also used their own experience as commuters.

Activity

Using your understanding of the issues associated with commuting, develop a problem statement to describe the problem that the design team should solve.

On the basis of the research that they performed, the design team defined their design problem to be “Design a commuter vehicle that is environmentally friendly, acceptable to a typical commuter, and compatible with existing transportation infrastructure.”

The design team also expanded this problem statement to make it more informative as follows. Environmentally friendly means that the vehicle produces as little pollution and greenhouse gases as possible and uses sources of **renewable energy**. Acceptable to the commuter means that the vehicle is convenient (does not require the commuter to wait), comfortable, and affordable. Compatible with existing infrastructure means that the vehicle does not require changing roads, bridges, etc., and does not require the development of a new fuel distribution system.

Identify Criteria and Constraints

The problem statement is used as a starting point to develop an understanding of the characteristics of a good solution. These characteristics are described in terms of constraints and criteria. A constraint is a limitation or condition that must be satisfied by a design. A criterion is a standard or attribute of a design that can be measured. The constraints and criteria are used in subsequent steps of the design process to determine which of many possible designs should be implemented.

Activity

Using your problem statement or the one developed by the design team, develop criteria and constraints that could be applied to decide whether a potential commuter vehicle design is good or not.

From the problem statement, the SCV design team identified criteria and constraints that would apply to their design. They identified the following constraint:

- Does not require new transportation infrastructure.

They identified the following criteria:

- The amount of pollution and greenhouse gases emitted per mile traveled by a commuter.
- The percent of the energy used from renewable sources.
- The convenience for the commuter.
- The comfort of the commuter.
- The cost to use the vehicle for five years (includes the purchase price, maintenance, and fuel).

Activity

Compare your criteria and constraints with the ones developed by the design team. What are the strengths of your criteria when compared to the design team's? What are the weaknesses?

Are each of your criteria measurable? Does each accurately reflect the problem statement?

Generate Concepts

With criteria and constraints identified, the design team begins to **generate concepts** for the design. This is the step in which creativity plays a very important role—good designs are often very different from existing solutions to a problem. In addition to creativity, the design team must use discipline to ensure that they explore enough options and potential solutions to guarantee a good design. Therefore, it is important to use a structured process to generate concepts for a design. Many different processes could be used. The one presented here is adapted and simplified from *Product Design and Development* by Eppinger and Ulrich. It includes the steps of problem decomposition, searching externally and internally for ideas, and systematically exploring possibilities.

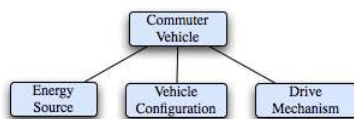
Decompose the Problem into Subproblems

When a design problem is complex, it can be very beneficial to **decompose** the problem into subproblems. Subproblems are smaller problems that must be solved in order to solve the overall problem.

Activity

Think of as many subproblems as you can for the SCV.

The SCV team broke the overall problem into subproblems as shown (Figure below). Each of the subproblems is simpler to approach than the whole problem. The energy source is how the vehicle gets energy to move; for example, the energy source for a regular car is gasoline. The vehicle **configuration** is the number of wheels on the vehicle and where they are placed relative to the driver. The drive mechanism transforms energy into the locomotion of the vehicle.



Decomposition of the commuter vehicle problem into three some problems: energy, configuration, and drive mechanism.

Search Externally for Ideas

Once the problem is decomposed into subproblems, the design team can begin to search for ideas to solve each subproblem. One source of ideas is to look at existing products and ideas to see whether there are already solutions to the overall problem or the identified subproblems. Sources of external information include interviews with potential customers or experts in the subproblem areas, patent and other technical databases, and existing products. Much of this information is now available on the World Wide Web.

Activity

Identify sources of information that you could use to find ideas for your subproblems. Use one these resources to develop a list of potential solutions to one of your subproblems.

The design team researched externally to find potential energy sources for their commuter vehicle. They discovered the following energy sources:

- Solar energy converted into electricity using **photovoltaic solar cells**. This is the power source used by the Mars rover Sojourner (Figure below).
- Nuclear energy
- Wind Energy converted into electricity using a turbine and generator. This might be a smaller version of a wind turbine such as that shown here (Figure below).
- Human power.
- Gasoline, a nonrenewable fossil fuel.
- A fuel cell that converts hydrogen and oxygen into electricity. The Figure below shows the fuel cell that provides power to the Toyota Fuel Cell Hybrid Vehicle (FCHV).
- Ethanol made from corn or other plants; ethanol can typically be used like gasoline with only slight modification to the car's fuel system.

After some reflection, the team discarded nuclear energy and a fuel cell as being **unfeasible** given the current state of technology.



The rover Sojourner on the surface of Mars. The flat black panel on the rover's top is a panel of photovoltaic solar cells that provided power for 83 days.



Large wind turbines on a wind farm in Iowa. These turbines use the wind's energy to generate electricity.



The Toyota Fuel Cell Hybrid Vehicle is powered by a fuel cell that generates electricity from hydrogen and oxygen. The fuel cell is still experimental technology that is currently extremely expensive, but shows promise for the future.

The team also searched for possible drive mechanisms. They settled on three types:

- A clutch, gearbox, and drive shaft similar to the drive train used for most manual transmission, rear wheel drive automobiles.
- Electric motors that are located in the hubs of each wheel and drive each wheel directly. The Figure below shows a bicycle hub that contains an electric motor; electric motors can also be put in the hubs of car wheels.
- A chain drive similar to that used in motorcycles and bicycles. Figure below shows a motorcycle chain drive.



This bicycle hub contains an electric motor that moves the bicycle.



This chain drive transmits power from the motorcycle engine to its rear wheel.

Search Internally for Ideas

Searching internally for ideas is often called brainstorming; The Figure below illustrates a group brainstorming. The goal of brainstorming is to develop as many ideas as possible without worrying whether they are feasible. Sketches are often good tools to capture ideas and to generate new ideas.



Students write their ideas on white boards during a brainstorming session.

Activity

Brainstorm ideas that could solve one of your subproblems.

To solve the vehicle configuration subproblem, the design team brainstormed several possible configurations; a configuration is an arrangement of wheels around the passenger compartment. They brainstormed four different configurations, each have between one and four wheels. After brainstorming the configurations, they found photographs online to represent each configuration. These photographs are presented in the Figure below. After reflection, the team discarded the one wheel configuration as being unfeasible. They noted that both two-wheel configurations would require some method of balancing, but kept them both because there are existing vehicles that use each configuration.



Possible solutions to the subproblem of configuring the wheels around the passenger.

Explore Systematically

Searching externally and internally will generate many possible solutions for each of the subproblems. To ensure that good solutions are not left out of the set of possible designs, it is important to use a structured process to examine possible combinations of subproblem solutions. A tool for systematic exploration is the concept combination table. In this table, solutions for each of the subproblems are combined; Figure below shows a concept combination table for the commuter vehicle.

Energy Source	Vehicle Configuration	Drive Mechanism
Solar cells (photovoltaic)	Four wheels	Clutch, gear box, and drive shaft
Human Power	Three wheels	Wheel hub motors
Wind Energy	Two wheels (bicycle)	Chain drive with gears
Gasoline (fossil fuel)	Two wheels (side to side)	
Ethanol		

The concept combination table for the commuter vehicle.

To use the table, a solution for each subproblem is combined, and then a sketch or description of the resulting concept is created. For example, if the concepts are combined as shown in the Figure below, then the possible design in Figure below results. This design could be very similar to a standard bicycle with an added solar cell canopy that shades the driver. The pedals of the bicycle would be removed and replaced by an electric motor that drives the vehicle forward.

Energy Source	Vehicle Configuration	Drive Mechanism
Solar cells (photovoltaic)	Four wheels	Clutch, gear box, and drive shaft
Human Power	Three wheels	Wheel hub motors
Wind Energy	Two wheels (bicycle)	Chain drive with gears
Gasoline (fossil fuel)	Two wheels (side to side)	
Ethanol		

The concept combination table is used to generate a particular possible design.



A sketch of the possible design obtained from the concept combination table above. Note that engineers often use rough, hand-drawn sketches at this point in the design process to understand design concepts and explore their strengths and weaknesses.

Note that the combination of design elements often does not provide a complete design concept; decisions must be made to fill in the gaps. For example, if solar cells are included as part of a design, they could be placed on the vehicle or they could be part of a fixed charging station that charges a battery on the vehicle; the design team must decide which configuration would make the most sense.

Activity

Using the solutions to subproblems that you have developed in previous activities, create a concept combination table for the problem. Use your concept combination table to generate five or six design concepts. Sketch each of your concepts.

The design team used the concept combination table to develop six concepts.

Concept 1: The design in the Figure above.

Concept 2: The energy source is a combination of solar cells and wind energy; the solar cells and wind turbines are installed at centrally located municipal charging stations and used to charge batteries. The vehicle configuration is a small, three-wheeled car with an enclosed passenger cabin that seats two people. The drive mechanism is wheel hub motors installed in the three wheels, with energy supplied to the motors from the charged batteries; these motors use **regenerative braking** to recover energy as the vehicle slows down.

Concept 3: This concept combines a gas engine with a four-wheeled configuration and a clutch, gearbox and drive shaft to form a traditional automobile. To be attractive as an alternative commuter vehicle, this design would be a two-seater subcompact.

Concept 4: This concept is the same as Concept 3, except that the engine is run on ethanol. Thus, Concept 4 is a small, two-seater alternative fuel vehicle.

Concept 5: The energy source is a combination of solar cells and human power. The vehicle configuration is three wheels, and the drive mechanism is a chain drive with gears. This is similar to a solar-assisted tricycle.

Concept 6: The energy source is a combination of solar cells mounted on the vehicle plus a battery; the battery can be charged at the user's home using a renewable energy source (wind or solar cells) or plugged into the user's home electricity system. The battery provides most of the energy, while the solar cells extend the life of the battery on sunny days. The configuration is two wheels side by side with room for a single passenger, and the drive mechanism is wheel hub motors. The motors are controlled to keep the vehicle balanced (similar to the Segway personal transport device in the Figure above).

Some combinations will not make sense or will result in a concept that is clearly unfeasible. For example, any concept that uses wheel hub motors must use an energy source that generates electricity.

Explore Possibilities and Select a Design

The design concepts are explored to understand their characteristics. For example, exploring Concept 1, the solar-powered bicycle in Figure above, leads to the following conclusions:

- The design would use only renewable energy.
- The design would be relatively inexpensive to manufacture and would cost nothing to operate.
- The design may not be convenient for the commuter, since the motor will only run when sunlight falls on the solar array. This means that it is impossible to commute at night or on cloudy days.
- The design will not be particularly comfortable for the commuter, since they will be exposed to hot, cold, and rainy weather, and the seat appears to be uncomfortable.

Activity

Explore the possibilities of one of the concept combinations developed in the previous activity.

When exploring the possibilities of a design concept, the team may discover ways in which the design can be improved. For example, Concept 1 might be improved by providing a more comfortable seat and by adding a battery that can store energy for use when it is dark or cloudy and the solar array does not generate electricity.

Once several design concepts have been developed and explored so that their advantages and disadvantages are understood, the design team must choose one concept that will be used to create the design for the product. It is usually best to choose the concept using a structured decision process. In a structured decision process, each of the concepts is evaluated to see whether it meets the constraints and is compared with the other concepts using the criteria; the best concept according to the criteria that meets the constraints is typically selected to implement the product.

In the case study, Concept 2 did not meet the constraint because it would require cities to build charging stations, so it was eliminated from consideration. Using the criteria as a guide, the design team determined that the two best designs were Concept 1 and Concept 5. They ranked high because of low pollution, using only renewable energy, and being low cost compared with other options. However, it is clear that these designs are the least comfortable for the consumer, and may therefore not be commercially successful. At this point, the design team could choose to use one of these designs and go forward in the design process; or, they may feel after seeing the outcome of the selection process that their criteria did not accurately capture their customers' desires. In this case, they may go back and improve their criteria, then repeat the decision process. Or, they may determine that better concepts may have been developed with different combinations of subproblem solutions or through different assumptions in the Explore Possibilities step, and thus repeat the Concept Generation and Explore Possibilities steps.

Sometimes, a design does not satisfy the constraints but could be easily modified to satisfy the constraints. For example, in Concept 2 if a battery charging station were to be built at each customer's house, the concept could be judged to meet the constraint. At other times, one design will score low because it has a particular flaw that can be corrected by combining it with characteristics of another design. Thus, the team should see if there are any designs that score low because of one aspect and can be corrected or if two designs can be combined to provide a better design.

Develop a Detailed Design

After concept selection, the team has a general design concept; they have decided how each subproblem will be addressed and have an overall understanding of the design. Before the design can be manufactured, the team needs to develop the details of the design. A detailed design includes

- The shapes and dimension of all physical components.
- An understanding of which components will be acquired from external vendors and which will be fabricated within the company and, if fabricated within the company, the materials and fabrication processes to be used.
- A detailed schematic diagram of any electrical subsystems and computer code for any embedded processors.
- Assembly processes.

The development of a detailed design from a design concept may occupy the majority of time allocated to a new product design project. This step will also have a significant impact on the success of the project; a poor detailed design can ruin a good design concept.

In the process of developing a detailed design, the team may use many or all of the subsequent design steps of prototyping, testing, and refinement. This process may require many iterations as the testing of prototypes reveals previously unknown characteristics of the design.

A major step in the process of going from a design concept to a detailed design is the development of the design architecture. The design architecture is “the assignment of the functional elements of the product to the physical building blocks of the product” (Eppinger and Ulrich, 2003).

For example, one architectural decision for the SCV design is how to incorporate the solar array into the design. Should the array be a separate physical block of the vehicle, for example creating the canopy structure in the Figure above, or should the array be created as an integral part of the frame? The first option represents a modular architecture, while the second option represents an integrated architecture.

Prototype, Test, and Refine

A prototype or model is a representation of some aspect of the design. The purpose of models and prototypes is to provide additional understanding of the design and its performance. A prototype may implement only a small portion of design or may be comprehensive and implement the whole design. For example, while developing a detailed design for Concept 1, the design team may initially wish to develop a prototype only of the electrical system (the solar cell array and the electric motor). Once the electrical system design is verified, they may implement a comprehensive prototype of the whole vehicle.

Prototypes may be physical or virtual. A physical prototype may be implemented out of materials that are very similar to those that will be used to manufacture the final design, or, to reduce cost or save time, the prototype may be implemented out of other materials. A virtual prototype may be created using a computer-aided design and drafting (**CADD**) program. Modern programs can simulate many aspects of a physical system, revealing flaws or promoting understanding of the design without the need to implement it physically.

One important function of a prototype is to test whether the design will work as expected. Understanding of the design and confidence that it will work is gained as prototypes are tested and evaluated relative to the constraints and criteria for the design. Testing procedures should be carefully planned to ensure that questions about the design are answered without requiring too much time and resources. The test results should be evaluated relative to specifications that reflect the constraints and criteria.

Testing and evaluation of the prototype may reveal weaknesses in the design or may provide information that can be used to improve the design. In this case, the design will often be refined, particularly if it does poorly with respect to some of the criteria or constraints. Sometimes, the chosen design concepts do not meet the criteria or constraints, and the design team must go back and perform more concept generation and then select another concept. This is an integral part of a spiral design process.

Communication and Implementation

As the design team has gone through the design process, they have kept records of the different processes that they used and results of these processes. Often, this information is used to create user manuals and maintenance manuals for the product. This information is important for team members who will be required to update or modify the design in the future. Lessons are learned in the design process that should be conveyed to other teams in the company or perhaps to external stakeholders in government or academia. An important part of the design process is to document these issues and communicate the results to the appropriate stakeholders.

As the design is completed, the effort to implement the design increases. If the design is of a product that is manufactured, a manufacturing system must be developed. For example, in the alternate commuter vehicle design, suppliers for components such as motors and solar cells must be located; facilities for manufacturing the frame are created; and a sales and marketing staff are identified.

Review Questions

Multiple Choice

The following questions will help you assess your understanding of the Discovering Engineering section. There may be one, two, three, or even four correct answers to each question. To demonstrate your understanding, you should find all of the correct answers.

1. Design problems are broken down into subproblems because
 1. each design team member needs a specific problem to solve
 2. the customer or stakeholders do not understand the overall problem
 3. smaller problems must be solved in order to solve the overall problem
 4. engineering companies make more money solving many smaller problems
2. When a design team searches externally for ideas they
 1. interview customers
 2. look at existing products
 3. look at technical databases
 4. talk to experts in the problem area
3. A concept combination table helps you to
 1. explore design ideas systematically
 2. see the complete design concept
 3. identify the overall design problem
 4. keep track of rejected designs
4. A concept screening matrix is used to
 1. select a design
 2. eliminate constraints
 3. develop a design
 4. eliminate criteria
5. A prototype can be
 1. a physical representation
 2. a scale model
 3. a virtual representation
 4. a final product
6. Implementation means that a
 1. physical model is built
 2. virtual model is built
 3. prototype is built
 4. product is manufactured
7. A design is refined because
 1. it has met the constraints and criteria

2. testing has found weaknesses in the design
 3. a product must go through the spiral design process
 4. there are no further improvements to make
8. Communicating processes and results is done by
1. posting designs on a website
 2. creating a users manual
 3. text messaging team members
 4. emailing manufacturers
9. A detailed design includes
1. a market analysis
 2. shapes and dimensions of all physical components
 3. computer code
 4. assembly process
10. The step in the design process called Explore Possibilities is used to
1. make additional designs
 2. improve the design
 3. understand the design characteristics
 4. test the prototype
11. Searching internally for ideas is called
1. mind searches
 2. design sessions
 3. idea dumps
 4. brainstorming
12. When engineers generate ideas in the design process they
1. use an unstructured approach
 2. use a step by step approach
 3. use a mathematical approach
 4. use a structured approach
13. Which techniques are used to define the design problem?
1. Find expert information
 2. Try to identify the real problem
 3. Gather information from customers
 4. None of the above

Free Response Questions

14. How can you tell the difference between a good design and a bad design?
15. What is the difference between engineering design and other types of design (architectural, fashion, etc.)?
16. How do you know that your design team has considered enough ideas to ensure that they develop a good design?
17. What are the characteristics of a good problem definition statement?
18. What are the steps of the design process? Why are they not always completed in order?
19. How do you use team decision-making tools in the design process?
20. How do you create a detailed design from a design concept?

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4.4: Introduction to Engineering Design Vocabulary

Artifact

An object made by a human being for a particular purpose.

CADD

CADD stands for computer-aided design and drafting. It is the practice of using computer software to represent the geometry of designed objects.

Carbon dioxide emissions

Carbon dioxide is a gas that results from burning fuels that contain carbon (e.g., coal and gasoline). Because carbon dioxide is a greenhouse gas that traps solar radiation, release of large amounts of carbon dioxide into the atmosphere by burning fossil fuels is believed to contribute to global warming.

Characterize

Something is characterized by discovering its distinctive features.

Concept generation

The process of developing ideas that may be used to create a design.

Configuration

An arrangement of the elements of a design in a particular form.

Constraint

A constraint is a limitation or condition that must be satisfied by a design. Constraints are either satisfied or they are not.

Criterion

A criterion is a measurable standard or attribute of a design; for example, weight and size are both criteria. Criteria are used to compare different possible designs and determine which better solve the design problem.

Customer

A person or organization that pays for the design either directly or through the purchase of a product.

Decompose

Decompose means to break down into simpler parts.

Design architecture

The design architecture is the assignment of the functions that the design performs to the physical building blocks of the design.

Dimension

A specification of height, width, depth, or length.

Greenhouse gas

A greenhouse gas is a gas in the atmosphere that traps solar radiation and re-radiates it as heat, leading to warming of the environment.

Incremental design

The incremental design process begins with an existing design that is modified.

Infrastructure

Infrastructure is the basic structures and organization needed for the operation of a society. For example, the automotive transportation infrastructure is the roads, bridges, traffic signals, traffic signs, etc. necessary to drive cars.

Iterative

An iterative process is one that may be repeated.

Model

A model is a purposeful abstract representation of some aspect of a design. Types of models include equations, physical representations, computer representations, and other representations.

Photovoltaic

Photovoltaic means that light energy is converted into electrical energy (see also solar cell).

Problem statement

A problem statement is a concise description of the problem or need a design will address.

Prototype

A prototype is a first or a preliminary model of the design or some aspect of the design. Prototypes are often physical models, but increasingly computer models are used as prototypes. Prototypes are used to evaluate designs and discover flaws and weaknesses.

Regenerative brakes

Brakes that slow a vehicle by converting its energy of motion into electrical energy that can be stored in a battery.

Renewable energy

Renewable energy is energy that comes from sources that are not permanently depleted by use. For example, solar and wind energy are renewable, while coal and oil are nonrenewable.

Root cause analysis

An analysis of a problem or situation to find the real cause (root cause) of the problem and deal with it; in the absence of a root cause analysis, people often deal only with the symptoms of the problem.

Solar cell

A solar cell is a device typically made of metal and semiconductors that converts light energy into electrical energy.

Stakeholder

A stakeholder is a person or organization who has a stake in (e.g. an interest in or one who may be affected by) a design project. Stakeholders often include users and customers, the design team, and the company that employs the design team.

Sustainability

A sustainable solution is one that can be continued without using up non-renewable resources.

Unfeasible

A design is unfeasible if it does not meet the constraints.

Viable

Viable means able to work successfully.

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CHAPTER OVERVIEW

5: Connecting Science and Mathematics to Engineering

- [5.1: About Connecting Science and Mathematics to Engineering](#)
- [5.2: Case History- How Math, Science, and Engineering Led to the First Pocket Radio](#)
- [5.3: What is the Role of Science and Mathematics in Engineering?](#)
- [5.4: How Do Math and Science Connect with Engineering in High School and College?](#)
- [5.5: Connecting Engineering Career Fields with Science and Engineering](#)
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5.1: About Connecting Science and Mathematics to Engineering

Engineering is the application of the principles of *mathematics* and *science* to the creation or modification of components, systems, and processes (which are often referred to as a product or an **artifact**) for the benefit of society. **Engineers** use a series of logical steps (the **engineering design process**) to create such artifacts which represent a balance between quality, performance, and cost. This chapter explores and examines the role and connections of math and **science** to engineering and the need to succeed in the study of those subjects for a professional career in engineering.

Chapter Learning Objectives

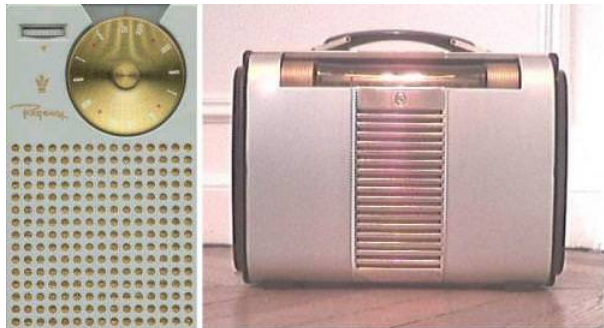
After working through this chapter, you should be able to do the following:

- Explain the goals and the nature of the fields of science, mathematics, and engineering and the differences between them.
- Explain generally how the fields of math and science and engineering benefit from one another as well as need one another.
- Explain what engineers in different disciplines do and the math and science they use.
- Explain the role of science and math in each step of the engineering design process.
- Describe the types of science and math that might be used by engineers in the different engineering disciplines along with an example for the design of a product or a process.

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5.2: Case History- How Math, Science, and Engineering Led to the First Pocket Radio

Imagine that it is November 1, 1954 and Dwight “Ike” Eisenhower is president and Leo Durocher’s Brooklyn Giants have just swept the World Series from the Cleveland Indians. Willie Mays has become a World Series legend after making “The Catch” in center field over his head with his back to the infield. Today, you have also just purchased a Regency TR-1 (Figure below), the world’s first “pocket” radio. It cost \$49.95 (equivalent to \$400 in 2007 dollars) with its four transistors, and you are now listening to Elvis Presley’s first hit, “That’s All Right”. The radio is gray, weighs 12 ounces, and with a size of 3”×5”×1”, you could slip it right into your pocket. This is a lot more convenient than the old vacuum tube portable radios which were bigger, bulkier, and heavier than the new transistor radio. One of these is an RCA 66BX, a six-tube portable radio that weighed 3 pounds and was 7”×5”×2” in size. Where did this incredible piece of shrinking technology come from? We shall see.



A portable transistor radio. The world's first transistor radio, the Regency TR-1 weighed 12 ounces with dimensions of 3" × 5" × 1"

The basic scientific knowledge necessary to develop a transistor radio started from the time when, on December 14, 1900, German physicist Max Planck explained to the world how an atom’s electrons behaved with a new theory called quantum mechanics. Over the next 20 years a **mathematical model** was developed for this theory, including an important equation called Schrödinger’s equation. From there, it was these basic principles of science and mathematics directed toward their practical application in electrical devices that led three researchers at Bell Labs on a race. The race was to **invent** a solid-state device that would replace bulky, unreliable, and energy-consuming vacuum tubes used in consumer electronics (such as radios) at the time. So it was that, on October 16, 1947, physicists John Bardeen, Walter Brattain and William Shockley, applied the mathematics and science of quantum physics to **semiconductors** to invent the world’s first transistor. They had created a device that could amplify a weak electronic signal 18 times over a wide range of frequencies. For their efforts they received the Nobel Prize in 1956.

Now that this new device existed, how would it be used? Texas Instruments used special materials processing techniques to make very pure semiconductor material necessary for transistors and started manufacturing them by 1952. Using those transistors, which cost \$2.50 each (\$2.50 will buy 100 million transistors on an **integrated circuit** today), engineers at the Regency Division of IDEA (Industrial Development Engineering Associates) of Indianapolis, Indiana, used the engineering design process to design, develop, and fabricate the world's first pocket radio. The electrical engineers at Regency used their industrial experience and the knowledge from their education on physics, mathematics, engineering science and electrical engineering to design a small radio; 100,000 units were manufactured. The connections of science and math to engineering are clear. The understanding of a **phenomenon** of the natural world, quantum physics, and the mathematical modeling of the phenomenon promoted the insights on the electrical behavior of semiconducting materials. It was the three researchers at Bell Labs who were searching for a solution to the well-defined problem of poor electrical behavior of vacuum tubes that led the team to invent the transistor. It was a very practical device indeed, since materials engineers at Texas Instruments were able to produce transistors in quantity so that another team of electrical engineers at IDEA could design, develop, and manufacture that first pocket radio.

The above case history of the first transistor radio illustrates the interplay of math, science, and engineering that occur in commercialization of a new device that later thrilled the country. And it did not take long for society to realize the benefits of quantum mechanics, Schrödinger’s equation, and the invention of the transistor. You could see it on some playgrounds a year after the Regency TR-1 was first merchandized. A crowd gathered around kids that brought their dads’ pocket radios to listen to the ‘55 World Series when the Brooklyn Dodgers beat the New Yankees in seven games. That miraculous little radio was a reflection of the “can do” spirit of the decade of the 1950s. Engineering could make science fiction reality. At the time cartoon detective character Dick Tracy had an imagined video-walkie-talkie-computer wrist watch, which he used to run down the city’s criminals. Part of it

became a reality but now, a half century later, today's high tech embodiment of Tracy's watch, the iPhone, can do everything Tracy's watch did and more. Next, let us consider the impact of technology in your own life from the exercise below.



An electric toothbrush and a microwave oven. The electric toothbrush was invented in 1939 but not popularized until the 1990s. The microwave oven was a spin-off of World War II radar technology and was invented in 1946.

Activity

How does technology affect you and your everyday life? Technology created by engineers affects everyone in their daily lives, usually in subtle understated ways. Try this exercise to explore the impact of technology with devices such as the electric toothbrush or the microwave, as shown in the Figure [above](#).

- Write down a short list of three or four electronic devices or gadgets that you use everyday.
- Write a short description of how you use them, how they affect your life, and how your life would be affected if they had not yet been invented.
- Take a guess at what kinds of engineers were involved in helping create one of the devices. Select one type of engineer and think about how she/he how might have used math and science in making the device?

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5.3: What is the Role of Science and Mathematics in Engineering?

This chapter has already introduced some ways in which science and math are connected to engineering. The chapter will continue to explore these connections in **invention**, **innovation**, education, careers, and design, as well as the impact on our daily lives. It is also becoming clear why it is critical to prepare for engineering education in college by taking and doing well in science and math courses throughout elementary, middle, and high school. In fact, the single best indicator of success in graduating with a college degree in engineering, science or math is taking courses in high school that include four years of math (at least through trigonometry) and three years of lab science. In the remainder of this chapter, we will now expand, articulate, detail, and exercise the engineering–math–science connection. The techniques of mathematics and the phenomena of science are like the brushes and colors on an artist’s palette. Just as an artist creates a new reality with his/her painting, so does an engineer create a new reality for how individuals live in a society. We have seen an example of this not only with the creation of the first pocket radio, but also how the reality of our daily lives have been impacted by other artifacts such as the cell phone and the computer. The next section will examine in greater detail the nature of the connection between math, science, and engineering.

What Is Engineering?

Engineering creates valued products such as the pocket radio. This is done by analyzing the nature of a problem or a need and then applying knowledge of math and science while completing the engineering design process to develop a solution to the design problem. A knowledge of science (e.g., chemistry, physics, and biology) helps the engineer understand the **constraints** inherent in a problem and helps the engineer develop possible approaches for a solution. Math (e.g., algebra, geometry, calculus, computer computation) is used both as a tool to create mathematical models that describe physical phenomena and as a tool to evaluate the merit of different possible solutions.

The profession of engineering is more formally defined by ABET, an organization that **accredits** college engineering programs, as “Engineering design is the process of devising a **system**, **component**, or process to meet desired needs. It is a decision-making process (often **iterative**), in which the basic sciences, mathematics, and the engineering sciences are applied to convert resources optimally to meet these stated needs.” Within this definition, science and mathematics are described as an essential part of the entire engineering process. They do not act alone within this process. “Engineers apply the principles of science and math to develop economical solutions to technical problems. Their work is the link between perceived social needs and commercial application.” (US Department of Labor) The goals of math and of science in engineering differ from those within the field of mathematics (where the goal is to quantitatively represent functional relationships) or the field of science (where the goal is to understand the natural world). In engineering, math and science are tools used within the engineering design process. Using the design process to address a problem or an issue leads to the solution of the problem and a product which might be a component, a system, or a process that fulfills a need that will benefit society. All fields of engineering use the tools of both math and science throughout all steps of the engineering design process. Effective use of math and science are critical to creating a high-quality solution to a need and an associated product of the process.

It often occurs that a high-quality product that comes out of an effective design process (including math and science) is taken for granted, such as a bridge standing for decades or centuries with no problems. Such a situation is illustrated with the Golden Gate Bridge, which has stood for more than seven decades since it was opened on May 27, 1937 (Figure below). When an inferior solution comes out of a flawed design process, catastrophe may result, and a bridge may collapse with possible loss of lives. Such an occurrence is illustrated below by the Tacoma Narrows Bridge Collapse (Figure below). On November 11, 1940, this bridge, located at Puget Sound, Washington, began swaying strongly in the wind and eventually broke up due to the resonance of the bridge which twisted until it broke up. There are mathematical tools available to analyze for waves and resonance of structures, but they were not applied in the design process of the bridge. This demonstrates the importance and need to broadly explore and utilize mathematics and science appropriate to the engineering design problem at hand.



The Golden Gate Bridge and the Tacoma Narrows Bridge. The Golden Gate Bridge, a signature landmark of San Francisco, had the world's longest span when opened May 27, 1937. The Tacoma Narrows Bridge, Puget Sound, Washington, broke up after in a strong wind on November 11, 1940.

Activity - Impact of disaster on an industry's design practice.

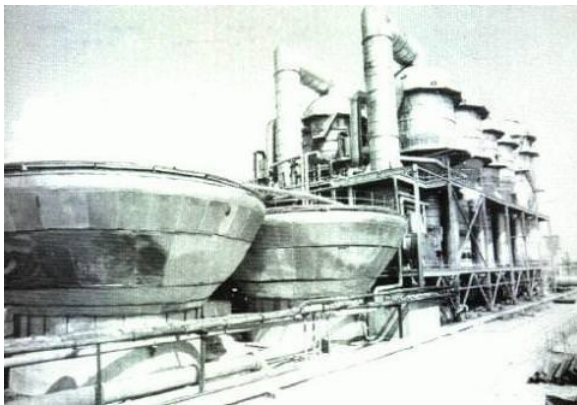
There have been many other disasters that occurred because of a flawed design process. One is the consecutive crashes of two British comet jets in 1954. Go to the two websites that describe the investigation and the resulting design changes: http://en.Wikipedia.org/wiki/De_Havilland_Comet and http://en.Wikipedia.org/wiki/TWA_Flight_800. How did the disasters cause engineers to change the airplane's design?

What Is Science?

The meaning of what science is can be debated and has changed over time. It is reasonable to think today of science as a process by which humans try to understand how the natural world works and how it came to be that way. The branches of science most frequently used by engineers include physics, chemistry, and biology. This is reflected by the fact that almost all undergraduate engineering programs require students to take foundational courses in those subjects. An example of how such science connects to engineering can be shown with the global problem of the lack of access to clean water by populations in some of the developing countries around the world. In developed countries turning on the bathroom faucet gives safe and drinkable water gushing from the tap. This safe and convenient water is actually a luxury that is not present everywhere. When it is not available, and water is not purified, people can become sick or even die from causes such as dehydration, cholera, dysentery, and cancer. So how have engineers figured out how to find, transport, purify, sanitize, and deliver water to those who need it? The following example should help answer that question as well as give a concrete example of how science connects with engineering.

Although more than half of the surface of the earth is covered with water, it is not accessible to plants, animals, or humans because of its 3–4% salt content. So, can we just remove the salt? No, because it is not so easy and can be costly. Science provides a variety of phenomena that can be used to desalinate water, and a few will be presented here. Chemistry tells us that there are many ways to desalinate water. One of the ways that water can be purified is by evaporation and condensation, which leaves the impurities in the original water. Another way is that, in freezing water the ice leaves the impurities behind in the unfrozen liquid. Another way is based on the fact that certain pore sizes of molecular membranes will allow water to diffuse through the membrane under pressure while leaving behind larger complexes of sodium and chlorine. Chemistry also has data on a variety of physical properties of water and salt water, including the thermodynamic data of heat of evaporation, heat of condensation, and heat of freezing. So what can be done with the different phenomena and all the data that are available?

The first question to answer is: Are there any types of engineering fields with individuals that work with such phenomena and data with the goal of desalinating water? The answer is yes. Chemical engineers are trained to use such phenomena and data to design and build large-scale processing plants for producing products such as cosmetics, antibiotics, and gasoline as well as purified water. Today, millions of gallons of water are being desalinated by flash evaporation, which uses evaporation and condensation phenomena, and by reverse osmosis, which uses molecular membrane technology based on membrane diffusion principles. A nuclear driven flash evaporation plant located on the Caspian Sea is shown in Figure below.



Nuclear powered flash distillation desalination equipment located on the Caspian Sea.

What Is Mathematics?

Math is utilized in a variety of ways within many fields of engineering. The most crucial aspects of math within this field involve cost–benefit–risk analysis and the use of mathematical models. “Mathematical models may include a set of rules and instructions that specifies precisely a series of steps to be taken, whether the steps are arithmetic, logical, or geometric. Sometimes even very simple rules and instructions can have consequences that are difficult to predict without actually carrying out the steps. ... Often, it is fairly easy to find a mathematical model that fits a phenomenon over a small range of conditions ... but it may not fit well over a wide range.” (AAAS) Given the importance of having an end result that not only works, but also is safe and dependable, it is easy to understand why the use of mathematical models is an invaluable aspect of the engineering design process, as well as how these models would be applied to various steps within this process. Nearly every aspect of mathematics can be, and is, applied to the engineering design process in some way. All fields within engineering require an advanced knowledge of, and the ability to properly use, many math skills, including (but not limited to) algebra, calculus, geometry, measurements, tables and graphic representations of results, mathematical formulas, and time lines.

Activity—What kinds of science and math do engineers need to address some of today’s global societal issues?

Consider the list of contemporary global societal issues shown in Table 1. From the list, select two or three global issues of interest to you and write them down. Think about them and then write about the math and science that might be used to address each of the particular issues.

Contemporary global societal issues
• Drought in the Southwest United States. • Lack of drinkable water in many nations. • Aging bridges that are prone to collapse. • Lack of accessible and reliable public transportation. • Need for renewable energy sources. • Children and others disabled by landmines and other weapons. • Environmental pollution from oil and chemical spills (Figure below). • Need for safer cars. • Global warming from carbon emissions from burning fossil fuels (Figure below). • Unsafe and unpleasant airports around the world.



Icebergs breaking off glaciers and an oil spill.

Review Questions

The following questions will help you assess your understanding of the What Is the Role of Science and Mathematics in Engineering? section. There may be one, two, three, or even four correct answers to each question. To demonstrate your understanding, you should find all of the correct answers.

1. Mathematical models

1. describe scientific phenomena
2. can evaluate cost

3. require a calculator
 4. are easier to build than physical models
2. Engineers use mathematics and science to
1. understand the world
 2. solve problems
 3. build models
 4. design a process
3. An example of a mathematical model is
1. the budget for product development
 2. drawings used to design a product
 3. the engineering design process
 4. rate of the flow of water through a hose

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